

THE MÉZARD–PARISI FORMULA FOR THE VIANA–BRAY MODEL

ABSTRACT. We prove the Mézard–Parisi variational formula for the Viana–Bray model with bounded symmetric couplings, a fixed external field, and arbitrary finite connectivity and inverse temperature. The variational class consists of finite Ruelle probability cascades with independent site marks. Diluted interpolation gives the upper bound. For the reverse bound, we construct a stationary spin law whose cavity value equals the limiting pressure, and minimize a Poissonized cavity functional while controlling its error uniformly over the full reweighting orbit. We then enlarge a minimizing law by the actual centres of its sampled clusters. Gaussian stationarity preserves these centres and the marked Ghirlanda–Guerra identities, while finite-helper approximations make the successive enlargements continuous. The resulting centre algebra yields conditional independence of site rows given their sampled genealogy. Consistent finite coarse genealogies then produce cascade trials converging in all finite replicated spin distributions and in cavity value. The cascade depth is allowed to grow throughout the approximation.

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1. INTRODUCTION AND MAIN RESULT

The Viana–Bray model is an Ising spin glass on a sparse random graph. Its cavity description associates a distribution of spin messages with each level of a hierarchical Gibbs state. The Mézard–Parisi functional evaluates the change in free energy produced by adding a vertex and subtracting the compensating edges. The central problem is to show that finite hierarchical trials approximate the limiting pressure at arbitrary temperature and connectivity.

Diluted interpolation bounds the pressure by every such trial. The reverse inequality requires a representation of enough of the limiting spin law to evaluate the entire cavity functional. This functional involves multioverlaps of all even orders, so recovering only the pair-overlap distribution does not provide the required approximation. Our proof constructs a minimizing stationary law together with an algebra of actual cluster centres. This additional structure determines the conditional joint law of its site rows and permits approximation by finite cascades.

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1.1. The model. Fix $\alpha, \beta \geq 0$, $h \in \mathbb{R}$, and a symmetric probability measure ν supported on $[-J_0, J_0]$, where $J_0 < \infty$. For each N , let $M_N \sim \text{Pois}(\alpha N)$. Let the couplings (J_k) have law ν , and let i_k, j_k be independent uniform indices in $\{1, \dots, N\}$. All these variables are independent. Define

$$(1.1) \quad Z_N = \sum_{\sigma \in \{-1, 1\}^N} \exp \left\{ \beta \sum_{k=1}^{M_N} J_k \sigma_{i_k} \sigma_{j_k} + h \sum_{i=1}^N \sigma_i \right\}, \quad F_N = \frac{1}{N} \mathbb{E} \log Z_N.$$

The field h is dimensionless. The independent ordered endpoints retain both self-loops and repeated edges. All parameters are fixed as N tends to infinity.

1.2. The cascade functional. We first specify the exact trial class. Fix an integer $r \geq 1$ and parameters

$$0 < \zeta_0 < \zeta_1 < \dots < \zeta_{r-1} < 1.$$

On the tree of finite sequences of positive integers of length at most r , put an independent Poisson process at each vertex b of depth $p < r$, with intensity $\zeta_p u^{-1-\zeta_p} du$. List its points in decreasing order as $(u_{bn})_{n \geq 1}$. For a leaf $a \in \mathbb{N}^r$, write $a|\ell$ for its ancestor at depth ℓ and set

$$(1.2) \quad w_a = \prod_{\ell=1}^r u_{a|\ell}, \quad v_a = \frac{w_a}{\sum_{b \in \mathbb{N}^r} w_b}.$$

The total mass is finite and positive almost surely, as proved in Lemma 2.2.

Let $f : [0, 1]^r \rightarrow [-1, 1]$ be measurable. For each message-copy index I , attach independent uniform marks U_b^I to the nonroot vertices, independently of the weights and of all other copies. The resulting message array is

$$m_a^I = f(U_{a|1}^I, \dots, U_{a|r}^I).$$

Thus different message copies have independent mark trees, while their messages are indexed by the same cascade leaves. Define

$$(1.3) \quad \psi_J(x, y) = \cosh(\beta J)(1 + \tanh(\beta J)xy), \quad x, y \in [-1, 1].$$

Let $D \sim \text{Pois}(2\alpha)$ and $K \sim \text{Pois}(\alpha)$. Using fresh couplings and distinct message copies, put

$$(1.4) \quad \begin{aligned} A_a &= \sum_{\varepsilon = \pm 1} e^{h\varepsilon} \prod_{k=1}^D \psi_{J_k}(\varepsilon, m_a^k), \\ B_a &= \prod_{k=1}^K \psi_{J'_k}(m_a^{k,1}, m_a^{k,2}). \end{aligned}$$

Within each expression, the counts and couplings are shared across leaves. The constructions of A and B are independent. The functional and its infimum are

$$(1.5) \quad \mathcal{P}(r, \zeta, f) = \mathbb{E} \log \sum_a v_a A_a - \mathbb{E} \log \sum_a v_a B_a, \quad \mathcal{P}_* = \inf_{r, \zeta, f} \mathcal{P}(r, \zeta, f),$$

where the infimum ranges over precisely the choices above. In particular, a trial uses a finite cascade, a deterministic measurable function f , and independent marks at the nonroot vertices of each message tree.

1.3. Main theorem.

Theorem 1.1. *For every choice of parameters and coupling law in (1.1),*

$$(1.6) \quad \lim_{N \rightarrow \infty} F_N = \inf_{\substack{r \geq 1, \ 0 < \zeta_0 < \dots < \zeta_{r-1} < 1 \\ f: [0, 1]^r \rightarrow [-1, 1] \text{ measurable}}} \mathcal{P}(r, \zeta, f).$$

The thermodynamic limit is finite and equals $\sup_{N \geq 1} F_N$. The cascade depths in the approximating trials may increase without bound.

1.4. Previous results and the remaining representation problem. Write $F = \lim_N F_N$ for the pressure in Theorem 1.1. The preceding work separates the formula into two questions: why every cascade trial bounds F , and how to find trials whose values approach F . The second question concerns the joint distribution of spins sampled from the Gibbs measure.

The interpolation bound. Panchenko and Talagrand [8] establish diluted interpolation bounds for hierarchical trials. In our normalization their argument gives

$$F \leq \mathcal{P}(r, \zeta, f) \quad \text{for every admissible trial,} \quad \text{hence} \quad F \leq \mathcal{P}_*.$$

Section 2 includes this argument and its exact remainder. The remainder involves products of spins from arbitrarily many replicas, where replicas are independent samples from the same Gibbs measure. The same information appears when the logarithmic cavity terms are expanded. For example, the four-replica quantity $N^{-1} \sum_i \sigma_i^1 \sigma_i^2 \sigma_i^3 \sigma_i^4$ enters in addition to the pair overlaps $N^{-1} \sum_i \sigma_i^1 \sigma_i^2$. Thus the reverse bound requires control of the joint spin observables used by the functional, including those involving more than two replicas.

A variational description through spin distributions. One can retain this information by recording the spins at every finite collection of sites in every finite collection of replicas. Panchenko's spin-distribution framework [4] takes this complete array law as the order parameter. Cavity invariance describes how the law behaves when fresh vertex or edge factors change the relative weights of the sampled configurations. Such identities lead to free-energy representations over classes of spin laws satisfying them.

The trial class in (1.5) imposes a more specific condition on these arrays. The sampled states have a common tree, and each site is generated from an independent copy of the marks on that tree. Accordingly, a variational description over spin laws leaves a representation problem: find a law with the right cavity value that can be approximated by arrays with precisely this independence across sites.

The passage from a hierarchy to a spin representation. Panchenko [5] proves hierarchical exchangeability of pure states under the relevant replica identities. This gives invariance under relabellings that preserve the hierarchy. The resulting representation still leaves a question about the joint spin moments: is their remaining shared randomness determined by that hierarchy? In later work [6], a suitably perturbed diluted Hamiltonian is shown to have the predicted spin representation whenever the limiting overlap takes finitely many values. For a general overlap distribution, the corresponding passage to finite levels must approximate the spin arrays well enough to preserve the cavity value.

The present proof carries out this passage for a law selected by an auxiliary minimization. Its central structural conclusion is that, conditional on a hierarchy constructed from the spin profiles and their cluster observables, the site rows are independent with a common conditional distribution. This conclusion supplies the finite-tree approximation required by the variational formula.

1.5. Strategy of the proof. Section 2 gives $F \leq \mathcal{P}_*$. For the reverse inequality, we must construct finite cascade trials with values arbitrarily close to F . We first work with a general random law of spin profiles, on which the two cavity terms can already be evaluated. The proof must then solve two problems: keep its cavity value close to F , and obtain the independence across sites required by the trial class.

1. Choose a law that can be modified without losing control of its value (Section 3). The finite-system cavity calculation supplies a spin law with cavity value F . We introduce auxiliary random weights with a parameter $0 < \theta < 1$ and minimize the resulting cavity functional P_θ . The class consists of laws unchanged by positive reweightings depending on finitely many coordinates, after those coordinates are deleted. Writing p_θ for this minimum and $c = \beta J_0$, we prove

$$(1.7) \quad p_\theta \leq F + \frac{2\alpha c(1 - \theta)}{\theta}.$$

It is now enough, at fixed θ , to approximate a suitable minimizer by finite trials.

The point of this choice is that further structure can be imposed at no cost. For a minimizing law ρ , let T_U multiply profile weights by a permitted positive factor U , normalize, and delete its coordinates. Applying U before or after a fresh vertex factor A gives the same total normalization. Since the law is unchanged by the A operation, its expected cavity normalization is unchanged by U . The same holds for B . Thus

$$P_\theta(T_U \rho) = P_\theta(\rho) = p_\theta.$$

Averaging repeated operations and taking limits preserves this equality. We can also condition on parameters preserved by these operations, when the resulting laws remain in the allowed class. Their cavity values are at least p_θ and average to p_θ , so almost every conditional law is again minimizing. This is how the construction makes the required moment parameters deterministic.

2. Record the information shared within each cluster (Sections 4–6). A tree describes which sampled profiles belong to the same clusters. It need not describe all the information shared by their spins. The proof therefore adds observables that will let us replace the descendants of a branch by information belonging to their common cluster.

For a bounded observable, regard its values at all sites as a profile. Its *cluster centre* is the point of minimum norm in the closed convex hull of these profiles within the cluster. The squared norm is the limiting average of squared coordinates over the sites. This geometric choice serves two purposes. It is preserved when positive weights change on the same set of profiles, and its inner products identify averages over descendants in different child clusters. Sections 4–5 verify that the centres recomputed from each transformed law agree with these preserved geometric objects. This is what makes the new reweightings in Step 1 legitimate.

The construction includes products of observables and their centres, then repeats these operations. To probe their inner products, we multiply weights by e^{tg_a} , where $t \in \mathbb{R}$ and the centred Gaussian variables g_a have an observable's profile inner products as covariances. Differentiating the expected logarithmic normalizations gives identities to which the overlap results in Section 6 apply. The resulting structure has two properties needed below. First, all these profile inner products are fixed functions of the common tree and its branch heights. Second, further profiles can be sampled in different children of an observed branch without changing the old spin observations conditional on their tree. This tree is constructed from the whole collection of observables.

3. Average over descendants and remove branches (Section 7). Here is the calculation that turns the preceding structure into independence. Fix a deepest branch and sample profiles $a_1, a_2, \dots$ in different children of that branch, using the second property above. Write X_i^a for the spin at site i in profile a . Fix two distinct sites i, j , and let b_i, b_j be the coordinates of the spin centre of this cluster. If Γ records the observed tree and its branch heights, the centre identities give

$$(1.8) \quad \mathbb{E} \left[\left(\frac{1}{m} \sum_{\ell=1}^m X_i^{a_\ell} X_j^{a_\ell} - b_i b_j \right)^2 \middle| \Gamma \right] \leq \frac{4}{m}.$$

To see the mechanism, expand the square. For two different descendants, the descendant–descendant, descendant–centre, and centre–centre inner products agree, so their contributions cancel. Only m diagonal terms remain, each bounded by 4, divided by m^2 .

The extended descendants form an exchangeable sequence, even with the outside observations and common centre coordinates retained. Conditional on its empirical law, they are independent samples from that law. The empirical average is therefore its two-site moment, which the estimate identifies as $b_i b_j$. Applying the same calculation to every finite product of bounded observables determines this empirical law from the common centre coordinates and shows that its moments factor across sites.

We can consequently replace the leaves at this branch by their centre observables. At each site the replacement is a product of centres. At the next branch we need the centre of that product,

which explains why Step 2 must repeat both operations. Continue upwards until only one profile remains. Its empirical moments for all the resulting observables have been made deterministic, with the branch heights fixed. Thus no further random parameter remains to couple its sites. Working back through the replacements proves the desired assertion: for n sampled profiles, put $Y_i = (X_i^{a^1}, \dots, X_i^{a^n})$ and let Γ_n record their tree and branch heights. Then

$$(1.9) \quad \text{Law}(Y_1, \dots, Y_d \mid \Gamma_n) = \text{Law}(Y_1 \mid \Gamma_n)^{\otimes d}.$$

This is the precise connection to independent site marks in a trial.

4. Approximate by finite trials (Section 8). Retain finitely many levels of the tree. Represent the conditional law of one site on this finite tree, take independent copies for the other sites, and choose independent cascade weights with the corresponding tree distribution. These are new trials. As more levels are retained, their conditional one-site laws approach those of the minimizing law. Equation (1.9) then gives joint convergence at every finite collection of sites and sampled profiles.

This convergence suffices for the cavity terms. Truncate the Poisson degrees, approximate the logarithm by a polynomial on the resulting bounded interval, and observe that each polynomial term uses only finitely many sampled spins. The degree tails have a uniform integrable bound. The resulting trial values tend to p_θ , so $\mathcal{P}_* \leq p_\theta$ and

$$(1.10) \quad F \leq \mathcal{P}_* \leq p_\theta \leq F + \frac{2\alpha c(1-\theta)}{\theta}.$$

All constructions are completed at fixed $\theta < 1$. Letting $\theta \uparrow 1$ gives $F = \mathcal{P}_*$.

2. THE CASCADE FUNCTIONAL AND THE UPPER BOUND

We first establish the analytic properties of the finite-cascade functional. The Poisson normalization gives both its integrability and the product identity needed at the endpoints of the diluted interpolation. The same interpolation then proves the upper bound, with an exact remainder involving every even multioverlap. We conclude with the approximation results that will be used when finite cascades are recovered from a stationary spin law.

Throughout this section, write

$$c = \beta J_0, \quad \tau = \tanh c < 1, \quad t_J = \tanh(\beta J), \quad c_n = \mathbb{E} t_J^{2n}.$$

2.1. Normalization and recursive evaluation.

Lemma 2.1. *For every $x, y \in [-1, 1]$,*

$$(2.1) \quad e^{-c} \leq \psi_J(x, y) \leq e^c.$$

Consequently, for the vertex and edge factors defining the trial functional,

$$(2.2) \quad 2 \cosh h e^{-cD} \leq A_a \leq 2 \cosh h e^{cD}, \quad e^{-cK} \leq B_a \leq e^{cK}.$$

These inequalities remain valid after integrating the factors against any probability measure on the leaves.

Proof. The identity

$$\psi_J(x, y) = \frac{1+xy}{2} e^{\beta J} + \frac{1-xy}{2} e^{-\beta J}$$

expresses the factor as a convex combination of two numbers in $[e^{-c}, e^c]$. Multiplying this bound over the factors in A_a and B_a , and then summing over the spin in A_a , proves (2.2). Integration preserves each bound. $\square$

Lemma 2.2. *The sum of the unnormalized cascade weights is almost surely finite and positive, and its logarithm is integrable. Let $V_a > 0$ be a common measurable function of the node marks along the path to a . Suppose that the marks are independent over vertices, identically distributed at each depth, and satisfy the deterministic bound $|\log V_a| \leq C$. Vector-valued marks are allowed. Along a single path, define recursively*

$$(2.3) \quad X_r = \log V, \quad X_p = \frac{1}{\zeta_p} \log \mathbb{E}_{p+1} e^{\zeta_p X_{p+1}}, \quad p = r-1, \dots, 0,$$

where $\mathbb{E}_{p+1}$ integrates the next mark conditional on its ancestors. Then

$$(2.4) \quad \mathbb{E} \log \sum_a v_a V_a = X_0.$$

If $V_a^1, \dots, V_a^s$ are built from independent families of node marks, then

$$(2.5) \quad \mathbb{E} \log \sum_a v_a \prod_{j=1}^s V_a^j = \sum_{j=1}^s \mathbb{E} \log \sum_a v_a V_a^j.$$

Both identities hold conditionally on shared external data when the required independence holds conditionally. An integrable conditional bound on the logarithms permits subsequent averaging over those data.

Proof. Let S_z be the sum of the points of a Poisson process on $(0, \infty)$ with intensity $zx^{-1-z} dx$, where $0 < z < 1$. The sum of the points below one has finite expectation, and the number of points above one is almost surely finite. The process has infinitely many positive points. Thus $0 < S_z < \infty$ almost surely. Its Laplace transform is

$$(2.6) \quad \mathbb{E} e^{-tS_z} = \exp\{-\Gamma(1-z)t^z\}.$$

For $0 < q < z$, the identity expressing x^q as a positive constant times $\int_0^\infty (1 - e^{-tx})t^{-q-1} dt$ and (2.6) imply $\mathbb{E} S_z^q < \infty$. The Laplace integral for x^{-q} gives finite negative moments of every positive order. In particular, $\mathbb{E} |\log S_z| < \infty$.

If the points u_n carry independent positive marks Y_n with $\mathbb{E} Y^z < \infty$, the Poisson mapping theorem gives the intensity $(\mathbb{E} Y^z)zx^{-1-z} dx$ for the marked points $u_n Y_n$. Their sum therefore has the same law as $(\mathbb{E} Y^z)^{1/z} S_z$. Apply this fact recursively from the bottom of the tree. A child subtree whose first exponent is ζ_{p+1} has total weight equal in law to a deterministic multiple of $S_{\zeta_{p+1}}$. Its ζ_p moment is finite because $\zeta_p < \zeta_{p+1}$. It can therefore be used as the mark in the Poisson process at depth p . This proves that the root sum is a deterministic multiple of S_{ζ_0} , with finite positive normalization and integrable logarithm.

Apply the same induction to the sum with factors V_a , conditioning at each stage on the ancestor marks. The weighted subtree total has the law of the unweighted subtree total multiplied by e^{X_p} . At a parent vertex, the moment of the unweighted child total is the same deterministic factor in the weighted and unweighted calculations. Its cancellation leaves

$$(\mathbb{E}_{p+1} e^{\zeta_p X_{p+1}})^{1/\zeta_p},$$

which is precisely the recursion in (2.3). The two unnormalized root sums consequently have laws differing by the deterministic multiplicative factor e^{X_0} . Subtracting their expected logarithms proves (2.4). This subtraction only uses their marginal laws.

For a product of independent mark constructions, the terminal logarithm is the sum of the terminal logarithms of the factors. At every step of (2.3), the conditional exponential expectation factors over the independent constructions. The recursively transformed values therefore add, proving (2.5). Conditioning first on the external data gives the conditional statements. The assumed logarithmic bound justifies the final expectation. $\square$

In particular, the two logarithms in the trial functional are integrable. After subtracting $\log(2 \cosh h)$ from the vertex logarithm, their absolute values are bounded by cD and cK , respectively.

2.2. The diluted interpolation. Fix N and a finite trial (r, ζ, f) . For $s \in [0, 1]$, introduce independent Poisson counts with means $\alpha N s$ for physical edges, $2\alpha(1-s)$ for vertex-message factors at each site, and $\alpha N s$ for pure message edges. All factors use fresh independent couplings and message-array copies. With these counts denoted by M_s , $D_{i,s}$, and K_s , set

$$(2.7) \quad \hat{Z}_s = \sum_a v_a \sum_{\sigma \in \{-1,1\}^N} e^{h \sum_i \sigma_i} \prod_{k \leq M_s} \psi_{J_k}(\sigma_{i_k}, \sigma_{j_k}) \prod_{i=1}^N \prod_{k \leq D_{i,s}} \psi_{J_{ik}}(\sigma_i, m_{ik}^a) \prod_{k \leq K_s} \psi_{J_k}(m_{k,1}^a, m_{k,2}^a),$$

$$\varphi_N(s) = \frac{1}{N} \mathbb{E} \log \hat{Z}_s.$$

Here $\psi_J(\sigma_i, \sigma_j) = e^{\beta J \sigma_i \sigma_j}$. Lemma 2.1 gives integrability throughout the interpolation. Let $\langle \cdot \rangle_s$ denote averaging over replicas (σ^b, a^b) from its joint Gibbs measure. For $\ell \geq 1$, define

$$(2.8) \quad R_{1 \dots \ell} = \frac{1}{N} \sum_{i=1}^N \prod_{b=1}^{\ell} \sigma_i^b, \quad Q_{1 \dots \ell}(a^1, \dots, a^\ell) = \mathbb{E}_m \prod_{b=1}^{\ell} m^{a^b}.$$

The expectation $\mathbb{E}_m$ uses a fresh full message-array copy, independent of the interpolating Gibbs measure. Repeated leaf labels are included in this definition.

Proposition 2.3. *For every finite trial and every N ,*

$$(2.9) \quad \mathcal{P}(r, \zeta, f) - F_N = \alpha \sum_{n \geq 1} \frac{c_n}{2n} \int_0^1 \mathbb{E} \langle (R_{1 \dots 2n} - Q_{1 \dots 2n})^2 \rangle_s ds \geq 0.$$

Proof. For $X \in [-1, 1]$ independent of a fresh coupling J , symmetry of the coupling law gives

$$(2.10) \quad \mathbb{E}_J \log \{ \cosh(\beta J) (1 + t_J X) \} = \mathbb{E} \log \cosh(\beta J) - \sum_{n \geq 1} \frac{c_n}{2n} X^{2n}.$$

The logarithmic series is uniformly absolutely convergent since $|t_J X| \leq \tau < 1$.

Poisson differentiation expresses $\varphi'_N(s)$ as a sum of expected changes caused by adding one factor. A physical edge contributes (2.10) with $X = \langle \sigma_i \sigma_j \rangle_s$. Replicating $2n$ times and averaging the independent uniform endpoints i, j gives $\langle R_{1 \dots 2n}^2 \rangle_s$. A vertex-message factor similarly gives $\langle R_{1 \dots 2n} Q_{1 \dots 2n} \rangle_s$, after averaging its site and fresh marks. A pure message edge gives $\langle Q_{1 \dots 2n}^2 \rangle_s$. Its two message copies are independent, including when replica leaf labels coincide.

After division by N , the derivatives of the three Poisson means are α , -2α , and α . The constant $\mathbb{E} \log \cosh(\beta J)$ therefore cancels, and

$$(2.11) \quad \varphi'_N(s) = -\alpha \sum_{n \geq 1} \frac{c_n}{2n} \mathbb{E} \langle (R_{1 \dots 2n} - Q_{1 \dots 2n})^2 \rangle_s.$$

Each added factor changes the logarithm by at most c , so the add-one formula follows by differentiation of the absolutely convergent Poisson series. Moreover, the absolute value of the right side of (2.11) is bounded uniformly in N and s by

$$4\alpha \sum_{n \geq 1} \frac{\tau^{2n}}{2n} = 4\alpha \log \cosh c.$$

This justifies termwise expectation and integration, as well as the integrated formula at the endpoints.

At $s = 0$, summing over the spins factorizes the partition function over sites. Conditional on the counts and couplings, the site factors use independent node marks. The product identity (2.5) consequently gives

$$\varphi_N(0) = \mathbb{E} \log \sum_a v_a A_a.$$

At $s = 1$, the physical partition function factors out of the leaf sum. Writing

$$b = \mathbb{E} \log \sum_a v_a \psi_J(m_1^a, m_2^a),$$

and applying the same product identity conditional on the number of pure message edges gives

$$\varphi_N(1) = F_N + \alpha b, \quad \mathbb{E} \log \sum_a v_a B_a = \alpha b.$$

Subtracting these endpoint values in the integrated form of (2.11) proves (2.9). $\square$

Every even multioverlap appears in the remainder. Their coefficients follow directly from the logarithmic expansion, so the upper bound also identifies which replicated information the recovery argument must preserve. The interpolation is the $p = 2$ case of [8, Theorem 4, equation (5.10)]: its parameters are $a = \cosh(\beta J)$ and $b = \tanh(\beta J)$, with single-spin factor σ . The required sign condition $\mathbb{E}(-b)^n \geq 0$ follows from symmetry, and $|b| < 1$ follows from bounded couplings. The coefficients αp and $\alpha(p - 1)$ become 2α and α . The deterministic field is allowed in that result. Lemma 2.2 supplies its recursive cascade evaluation in the present normalization.

2.3. Thermodynamic convergence and parameter dependence.

Proposition 2.4. *The sequence $a_N = \mathbb{E} \log Z_N$ is superadditive and satisfies*

$$(2.12) \quad |a_N - N \log(2 \cosh h)| \leq \alpha N c.$$

Consequently,

$$(2.13) \quad F = \lim_{N \rightarrow \infty} \frac{a_N}{N} = \sup_{N \geq 1} \frac{a_N}{N}$$

exists and is finite. In particular, $F \leq \mathcal{P}_$.*

Proof. Split the sites into blocks of sizes N_1, N_2 , where $N = N_1 + N_2$, and put $\lambda = N_1/N$. Interpolate between independent models on the blocks and the full model by taking the mean number of full-system edges to be $\alpha N s$, and the mean number of edges within block b to be $\alpha N_b(1 - s)$. The field remains fixed. Let $\Psi(s)$ be the expected unnormalized log partition function. If $R_{1 \dots 2n}^{(b)}$ is the multioverlap restricted to block b , then

$$R_{1 \dots 2n} = \lambda R_{1 \dots 2n}^{(1)} + (1 - \lambda) R_{1 \dots 2n}^{(2)}.$$

Applying (2.10) to the three families of edges gives

$$\Psi'(s) = \alpha N \lambda (1 - \lambda) \sum_{n \geq 1} \frac{c_n}{2n} \mathbb{E} \langle (R_{1 \dots 2n}^{(1)} - R_{1 \dots 2n}^{(2)})^2 \rangle_s \geq 0.$$

The preceding domination justifies integration. The endpoint values are $a_{N_1} + a_{N_2}$ and a_N , so $a_N \geq a_{N_1} + a_{N_2}$. Comparing the partition function with the one having no edges by (2.1) gives (2.12).

For completeness, fix $m \geq 1$ and write $N = km + j$, with $0 \leq j < m$ and $a_0 = 0$. Superadditivity yields $a_N \geq k a_m + a_j$, hence $\liminf_N a_N/N \geq a_m/m$. Taking the supremum over m , and using the definition of that supremum for the reverse upper bound, proves (2.13). Proposition 2.3 then implies $F \leq \mathcal{P}(r, \zeta, f)$ for every trial, and taking the infimum proves the last assertion. $\square$

The formulas include all parameter choices in the theorem. If $\alpha = 0$, $\beta = 0$, or $\nu = \delta_0$, direct evaluation gives $F_N = \mathcal{P}(r, \zeta, f) = \log(2 \cosh h)$ for every trial. The same arguments apply at $h = 0$. For the zero-message trial,

$$(2.14) \quad \mathcal{P}(r, \zeta, 0) = \log(2 \cosh h) + \alpha \mathbb{E} \log \cosh(\beta J).$$

This also checks the net contribution of the hyperbolic-cosine factors.

Independent ordered endpoints have been retained in each interpolation. Thus self-loops and repeated edges are already included in the overlap identities. For example, when $N = 1$, every edge is a loop and

$$\log Z_1 = \log(2 \cosh h) + \beta \sum_{k \leq M_1} J_k, \quad F_1 = \log(2 \cosh h).$$

For $f = 0$, one has $Q_{1\dots 2n} = 0$ and $R_{1\dots 2n}^2 = 1$. Formula (2.9) therefore becomes

$$\mathcal{P} - F_1 = \alpha \sum_{n \geq 1} \frac{c_n}{2n} = \alpha \mathbb{E} \log \cosh(\beta J),$$

in agreement with (2.14).

For a fixed coupling law supported in $[-J_0, J_0]$, the finite-volume pressures also satisfy

$$(2.15) \quad |F_N(\alpha, \beta, h) - F_N(\tilde{\alpha}, \tilde{\beta}, \tilde{h})| \leq \beta J_0 |\alpha - \tilde{\alpha}| + \tilde{\alpha} J_0 |\beta - \tilde{\beta}| + |h - \tilde{h}|.$$

Indeed, vary the parameters in the order α, β, h . Poisson thinning and the add-one bound give the first term. Differentiating the finite-volume logarithm with respect to β and h gives bounds equal to the expected number of edges times J_0 and to N , respectively. The same estimate passes to F . The construction of the reverse inequality below will be made at each fixed parameter choice.

2.4. Root marks and approximation of message functions. For later use, temporarily enlarge the finite trial class by allowing a root mark at each site:

$$(2.16) \quad m_I^a = g(U_*^I, U_{a|1}^I, \dots, U_{a|r}^I).$$

For each message copy I , the mark U_*^I is shared across leaves. The root marks are independent for distinct copies and independent of all other marks and weights. Denote the resulting functional by $\mathcal{P}^{\text{root}}$.

Proposition 2.5. *For every extended trial of the form (2.16) and $0 < \eta < \zeta_0$, there is a trial in the specified class, with parameters $(\eta, \zeta_0, \dots, \zeta_{r-1})$, whose value $\mathcal{P}_\eta$ satisfies*

$$(2.17) \quad |\mathcal{P}_\eta - \mathcal{P}^{\text{root}}| \leq \frac{\eta c^2}{2} (5\alpha^2 + 3\alpha).$$

Consequently, adjoining independent site-root marks does not change the infimum $\mathcal{P}_*$.

Proof. Condition on all counts and couplings. Apply the cascade recursion to $\log A$ with the root-site marks fixed, and call its value X_A . The recursion preserves deterministic lower and upper bounds, so

$$X_A \in [\log(2 \cosh h) - cD, \log(2 \cosh h) + cD].$$

The conditional extended vertex functional is $\mathbb{E}X_A$, with this expectation taken over the root-site marks. Prepend a level with mass parameter η and use its node marks as the former root-site marks. The conditional vertex value of the resulting trial is $\eta^{-1} \log \mathbb{E} e^{\eta X_A}$. The edge term has the same description with a variable $X_B \in [-cK, cK]$.

If X lies in an interval of width $2b$, let $L(t) = \log \mathbb{E} e^{tX}$. Under every exponential tilt, $\text{Var}_t(X) \leq b^2$. To see this, for an interval $[a, d]$ the inequality $\mathbb{E}[(X - a)(d - X)] \geq 0$ gives

$$\text{Var}(X) \leq (\mathbb{E}X - a)(d - \mathbb{E}X) \leq \frac{(d - a)^2}{4},$$

and the same reasoning applies to each tilted law. Hence $L''(t) \leq b^2$. Jensen's inequality and integration of this second-derivative bound give

$$0 \leq \eta^{-1} \log \mathbb{E} e^{\eta X} - \mathbb{E} X \leq \frac{\eta b^2}{2}.$$

Apply this separately to X_A and X_B , and average the counts. Since

$$\mathbb{E} D^2 = 4\alpha^2 + 2\alpha, \quad \mathbb{E} K^2 = \alpha^2 + \alpha,$$

the sum of the two errors is the right side of (2.17). Every trial in the specified class is an extended trial whose rule ignores the root mark. Conversely, the displayed bound approximates every extended value by specified-class values as $\eta \downarrow 0$. The two infima are equal. $\square$

If the message rule additionally depends on a single global environment parameter, with the conditional construction being an extended trial for almost every environment value, its conditional functional can be disintegrated over that parameter. The average of the conditional values is at least their infimum. Thus one may fix such a global parameter when seeking an upper bound on $\mathcal{P}_*$. Marks attached to individual states or clusters remain part of the conditional trial.

Lemma 2.6. *For fixed r, ζ and measurable $f, g : [0, 1]^r \rightarrow [-1, 1]$, put $\delta = \int_{[0, 1]^r} |f - g|$. Then*

$$(2.18) \quad |\mathcal{P}(r, \zeta, f) - \mathcal{P}(r, \zeta, g)| \leq C_{\alpha, c} \delta, \quad C_{\alpha, c} = \sinh(c) \{ \mathbb{E} D e^{2cD} + 2\mathbb{E} K e^{2cK} \} < \infty.$$

The constant is independent of the depth and the mass parameters.

Proof. Couple the trials using the same weights and marks. For a vertex-message factor, (2.1) and the definition of ψ_J give

$$|\psi_J(\varepsilon, x) - \psi_J(\varepsilon, y)| \leq \sinh(c) |x - y|.$$

Conditional on $D = d$, telescope the d factors in A_a . Divide the resulting difference of leaf integrals by their common lower bound from (2.2), using $|\log x - \log y| \leq |x - y| / \min(x, y)$. After discarding a factor $e^{-c} \leq 1$, this gives

$$\left| \log \sum_a v_a A_a(f) - \log \sum_a v_a A_a(g) \right| \leq \sinh(c) e^{2cd} \sum_{k \leq d} \sum_a v_a |m_k^a(f) - m_k^a(g)|.$$

The marks are independent of the weights, and each leaf has the same uniform path-mark distribution. The expectation of the double sum is therefore $d\delta$.

For an edge factor, the inequality $|xy - x'y'| \leq |x - x'| + |y - y'|$ gives the same estimate with $2k\delta$ in place of $d\delta$, conditional on $K = k$. Averaging the two counts proves (2.18). Finally, for $P \sim \text{Pois}(\lambda)$,

$$\mathbb{E} P e^{2cP} = \lambda e^{2c} \exp\{\lambda(e^{2c} - 1)\},$$

which proves the asserted finiteness. $\square$

The conditional expectations of f on dyadic partitions of $[0, 1]^r$ remain in $[-1, 1]$ and converge to f in L^1 . Thus Lemma 2.6 permits such approximations for any fixed trial. To approximate a general stationary message law, we will also need continuity under convergence of full replicated arrays.

2.5. Continuity of the cavity functional. Consider a random probability measure G on a state space. Conditional on its environment, let $m_i(a) \in [-1, 1]$ be independent identically distributed random site functions. Independently of these functions, sample $a^1, a^2, \dots$ from G , conditional on the same environment, and set

$$X_i^\ell = m_i(a^\ell).$$

Using the vertex and edge factors of the trial functional, define

$$C = \mathbb{E} \log \int A_a G(da) - \mathbb{E} \log \int B_a G(da).$$

The integration against G replaces the weighted leaf sum. In particular, the message copies used in each factor are independent conditional on the environment.

Proposition 2.7. *Suppose that, along a sequence of such constructions, the law of every finite rectangle $(X_i^\ell)_{i \leq L, \ell \leq p}$ converges weakly to the corresponding law of another such construction. Then their cavity functionals C converge.*

Proof. Normalize the vertex integral by $2 \cosh h$. Conditional on $D = d$ and $K = k$, the normalized vertex integral and the edge integral belong to $[e^{-cd}, e^{cd}]$ and $[e^{-ck}, e^{ck}]$, respectively. Fix a degree cutoff L . Put $b = e^{cL}$ and $\vartheta = 1 - e^{-2cL}$. On $[e^{-cL}, e^{cL}]$, the polynomial

$$p_M(z) = \log b - \sum_{j=1}^M \frac{(1 - z/b)^j}{j}$$

approximates the logarithm with error at most

$$(2.19) \quad \frac{\vartheta^{M+1}}{(M+1)(1-\vartheta)}.$$

Indeed, $0 \leq 1 - z/b \leq \vartheta$, and the bound follows by summing the tail of the logarithmic series.

Each power of a G -integral is an expectation over independent replicas from G . Thus, for $d, k \leq L$, every polynomial expectation above is a bounded continuous function of a message rectangle involving at most $2L$ sites and M replicas, after averaging the independent couplings. Weak convergence of these rectangles gives convergence of the polynomial expectations. Letting M increase proves convergence of the logarithmic expectations at the fixed degree cutoff.

After subtracting the constant $\log(2 \cosh h)$, the omitted contribution is bounded uniformly over all the constructions by

$$(2.20) \quad c\mathbb{E}[D\mathbf{1}_{\{D>L\}}] + c\mathbb{E}[K\mathbf{1}_{\{K>L\}}].$$

Both terms tend to zero as $L \rightarrow \infty$. This proves convergence of the functionals and the required uniform integrability.

An explicit diagonal choice is available when $c > 0$. Take $L_n = n$ and $M_n = \lceil e^{2cn}(n + 2cn) \rceil$. The inequality $(1 - u)^m \leq e^{-um}$ shows that (2.19) is at most $e^{-n}/(M_n + 1)$. After fixing these two parameters, choose the array index large enough that the errors in the finite collection of polynomial expectations are at most $1/n$. The polynomial, array, and degree-tail errors all vanish. If $c = 0$, the cavity functional is the constant $\log(2 \cosh h)$. $\square$

The structural construction below will produce the finite-rectangle convergence in Proposition 2.7. This convergence retains the multioverlaps occurring in (2.9) and the higher replicated moments used to evaluate the logarithms.

3. STATIONARY SPIN LAWS AND A VARIATIONAL MINIMUM

The reverse inequality begins with a law of raw spins whose cavity value is the thermodynamic pressure. We first construct this law by a subextensive perturbation of the finite system. We then introduce Poisson–Dirichlet weights with a fixed exponent $0 < \theta < 1$ and minimize the resulting cavity functional over laws invariant under finite-coordinate reweightings. The comparison at the end of the section controls this minimum uniformly over the entire reweighting orbit. This uniformity will allow us to impose the additional structure needed for finite-cascade recovery.

3.1. Directing measures and ordinary reweighting. Put $\mathsf{X} = \{-1, 1\}$ and $E = \mathsf{X}^{\mathbb{N}}$, with the compact product topology. Let $\mathcal{C}$ be the set of probability laws of random Borel probability measures Q on E whose distributions are invariant under finite permutations of the coordinates of E . Thus $\mathcal{C}$ is a compact convex subset of $\mathcal{P}(\mathcal{P}(E))$. Conditional on Q , sample independent columns

$X^a = (X_i^a)_{i \geq 1}$ with law Q . The resulting array is exchangeable separately in its row and column indices. Conversely, the moments

$$\mathbb{E} \prod_{a=1}^k Q(g_a),$$

where the g_a are continuous cylinder functions, determine the law of Q . This identifies its weak topology with convergence of the finite replicated spin arrays. The same topological facts hold for any compact $\mathbf{X} \subseteq [-1, 1]$.

A positive cylinder factor is a continuous function $V : E \rightarrow (0, \infty)$ depending on finitely many coordinates. By exchangeability we may take these to be the first d coordinates. Write sh_d for their deletion and define

$$(3.1) \quad S_V Q = (\text{sh}_d)_\# \frac{VQ}{Q(V)}, \quad \ell_V(\rho) = \mathbb{E}_\rho \log Q(V).$$

We use the same notation S_V for the induced map on laws. When V has independent external randomness, that randomness is averaged in both expressions. Successive factors use disjoint fresh coordinate blocks. Finite-coordinate reweightings are continuous and affine on $\mathcal{C}$, and they commute on exchangeable laws. The normalizations satisfy

$$(3.2) \quad \ell_{UV}(\rho) = \ell_V(\rho) + \ell_U(S_V \rho).$$

Indeed, successive updates multiply the density by UV , and the two normalizing constants multiply to $Q(UV)$. Interchanging the two probe blocks does not change the input law. Continuity follows by testing the output against cylinder functions and using the positive lower bound on V .

On raw spins, the factors from the introduction reduce to

$$(3.3) \quad A(x) = 2 \cosh \left(h + \beta \sum_{j=1}^D J_j x_j \right), \quad B(x) = \exp \left(\beta \sum_{j=1}^K J'_j x_{2j-1} x_{2j} \right),$$

where $D \sim \text{Pois}(2\alpha)$ and $K \sim \text{Pois}(\alpha)$. We retain the notation S_A, S_B for the corresponding averaged transitions and put

$$C_1(\rho) = \ell_A(\rho) - \ell_B(\rho).$$

The bounds in Lemma 2.1 give

$$(3.4) \quad |\log Q(A) - \log(2 \cosh h)| \leq cD, \quad |\log Q(B)| \leq cK.$$

Consequently all these quantities are integrable, and degree truncation extends the preceding continuity and cocycle statements to A and B .

3.2. A stationary law with the thermodynamic cavity value.

Proposition 3.1. *There exists $\rho \in \mathcal{C}$ such that*

$$S_A \rho = \rho, \quad S_B \rho = \rho, \quad C_1(\rho) = F.$$

Proof. If $\alpha = 0$, the factors are the constants $A = 2 \cosh h$ and $B = 1$, so any coordinate-exchangeable law has the asserted properties. Suppose $\alpha > 0$, and write $a_N = \mathbb{E} \log Z_N$. Set $s_N = N^{1/3}$. To the exponent in the finite system add an independent $\text{Pois}(s_N)$ number of cavity stars, each having logarithm

$$\log A_N^{\text{raw}}(\sigma) = \log \sum_{\varepsilon = \pm 1} \exp \left(h\varepsilon + \beta \sum_{j=1}^D J_j \varepsilon \sigma_{i_j} \right).$$

Here $D \sim \text{Pois}(2\alpha)$ and the endpoints i_j are independent uniform sites in $\{1, \dots, N\}$. Let Q_N^* be the perturbed Gibbs measure and a_N^* its expected log partition function. With

$$C_A = \log(2 \cosh h) + 2\alpha c,$$

the uniform bound on a star logarithm gives

$$(3.5) \quad |a_N^* - a_N| \leq C_A s_N, \quad \frac{a_N^*}{N} \longrightarrow F.$$

We first obtain invariance under adding cavity factors. If $P \sim \text{Pois}(\lambda)$, then

$$(3.6) \quad d_{\text{TV}}(\text{Law}(P+1), \text{Law}(P)) = \frac{1}{2} \mathbb{E} \left| \frac{P}{\lambda} - 1 \right| \leq \frac{1}{2\sqrt{\lambda}}.$$

Adding one independent cavity star therefore changes the full quenched law of Q_N^* by total variation at most $1/(2\sqrt{s_N})$. Adding one independent original edge has the same effect with $\lambda = \alpha N$. These estimates apply in particular to every bounded replicated-spin cylinder test. After truncating D , the chance of a collision among the probe endpoints and m observed sites is $O((D+m)^2/N)$. Finite-site exchangeability then identifies addition of a star with S_A , and addition of an edge with the transition having factor $\exp(\beta J x_1 x_2)$, on any fixed set of retained coordinates. All errors vanish uniformly for $L \leq N \leq 2L$. Invariance under the single-edge transition also gives invariance under an independent $\text{Pois}(\alpha)$ number of its iterates, which is S_B .

To identify the cavity value, split the edges of the size- $N+1$ graph according to whether their endpoints are old or new. The old–old, new–old, and new–new counts are independent Poisson variables with means

$$\frac{\alpha N^2}{N+1}, \quad \frac{2\alpha N}{N+1}, \quad \frac{\alpha}{N+1},$$

respectively. The size- N edge process is obtained from the old–old process by adding an independent compensating process of mean $\alpha N/(N+1)$.

The star perturbations can be coupled at the two sizes with a vanishing error. Replace every perturbing endpoint equal to the new site by an independent uniform old site. Changing one endpoint changes the star logarithm by at most $2c$. The expected number of such endpoints is $2\alpha s_{N+1}/(N+1)$. Couple the numbers of stars by retaining their common Poisson part. Since the expected cost of an extra star is at most C_A , the expected uniform Hamiltonian error is bounded by

$$(3.7) \quad \frac{4\alpha c s_{N+1}}{N+1} + C_A |s_{N+1} - s_N| = o(1).$$

The new–new loop contributes at most $\alpha c/(N+1)$ in expected absolute value.

Let Q_N^- be the Gibbs measure of the old–old graph, with the size- N star perturbation. The two independent additions to this common base give

$$(3.8) \quad a_{N+1}^* - a_N^* = \mathbb{E} \log Q_N^-(A_N) - \mathbb{E} \log Q_N^-(B_N) + o(1),$$

where the Poisson means in A_N, B_N are respectively $2\alpha N/(N+1)$ and $\alpha N/(N+1)$. Replacing them by 2α and α costs $O(N^{-1})$. This follows by coupling the Poisson counts, since each added interaction changes the corresponding logarithm by at most c .

The old–old edge mean differs from αN by a bounded amount. Differentiating the Poisson probabilities with respect to their mean, and applying the same estimate as in (3.6), bounds the total variation distance of these two edge-count laws by the integral of $1/(2\sqrt{\lambda})$ between their means. This distance tends to zero. For bounded cavity degrees the logarithms in (3.8) are bounded, so Q_N^- can be replaced by Q_N^* . The envelopes (3.4) remove the degree restriction uniformly in N . The collision estimate above also replaces fresh uniform endpoints by distinct coordinate probes. Thus the right side of (3.8) is $C_1(\text{Law}(Q_N^*)) + o(1)$, with the finite-coordinate interpretation just described.

Pad configurations by $+1$ beyond site N and regard Q_N^* as a random measure on E . Average its laws over $N = L, \dots, 2L-1$. Compactness gives a weakly convergent subsequence, with limit ρ . Every fixed finite permutation acts within the first N sites for all sufficiently large N , so $\rho \in \mathcal{C}$. Continuity for fixed cylinder factors, followed by degree truncation, passes the stationarity estimates to the limit and gives $S_A \rho = S_B \rho = \rho$.

Finally, continuity of the truncated logarithmic normalizations and (3.4) yield

$$\begin{aligned} C_1(\rho) &= \lim_{L \rightarrow \infty} \frac{1}{L} \sum_{N=L}^{2L-1} (a_{N+1}^* - a_N^*) \\ &= \lim_{L \rightarrow \infty} \frac{a_{2L}^* - a_L^*}{L} = F, \end{aligned}$$

along the chosen subsequence. The last equality follows from (3.5). Averaging the increments before taking the limit supplies the required cavity value without requiring convergence of the individual increments. $\square$

3.3. Poisson–Dirichlet reweighting and its normalization. Fix $0 < \theta < 1$. Given Q with law $\rho \in \mathcal{C}$, sample the columns X^a conditionally independently with law Q , and independently sample decreasing weights $(v_a)_{a \geq 1}$ with law $\text{PD}(\theta, 0)$. Gibbs brackets sample column labels independently with probabilities (v_a) . All expectations below include the directing law, the weights, and the external randomness.

Multiplying each weight by $V(X^a)$, normalizing, ranking again, and deleting the probe coordinates induces a different operation from (3.1). Its directing measure is

$$(3.9) \quad Q^V = (\text{sh}_d)_\# \frac{V^\theta Q}{Q(V^\theta)}, \quad T_V \rho = \text{Law}(Q^V).$$

If the factor also uses independent column-specific marks η_a , its likelihood in this formula is $\mathbb{E}_\eta V(x, \eta)^\theta$. Shared external randomness is conditioned on first and averaged afterward.

Lemma 3.2 (Poisson marking and normalization). *After reweighting, the weights have law $\text{PD}(\theta, 0)$ and are independent of the ranked columns, which are conditionally iid with directing measure Q^V . The maps T_V are continuous and affine on $\mathcal{C}$. For disjoint fresh probe blocks they commute, and*

$$(3.10) \quad \psi_V^{(\theta)}(\rho) := \mathbb{E}_\rho \log \sum_a v_a V(X^a) = \frac{1}{\theta} \mathbb{E}_\rho \log Q(V^\theta).$$

These continuous affine functionals satisfy

$$(3.11) \quad \psi_{UV}^{(\theta)}(\rho) = \psi_V^{(\theta)}(\rho) + \psi_U^{(\theta)}(T_V \rho).$$

The same statements hold with independent column-specific marks, using their θ -moment likelihoods.

Proof. Write $v_a = u_a / \sum_b u_b$, where the u_a are the points of a Poisson process on $(0, \infty)$ with intensity $\theta u^{-1-\theta} du$. Conditional on Q , mark the points independently with $x \sim Q$. Under $(u, x) \mapsto (uV(x), x)$, the intensity becomes

$$\theta u^{-1-\theta} du V(x)^\theta Q(dx).$$

Its first marginal is the stable intensity multiplied by $Q(V^\theta)$, and, conditionally on the transformed points, its marks are independent with law $V^\theta Q / Q(V^\theta)$. The normalized weights have law $\text{PD}(\theta, 0)$, independently of these marks. Deleting the probe rows gives (3.9).

The transformed stable sum has the same distribution as the original sum multiplied by $Q(V^\theta)^{1/\theta}$. The logarithms of both sums are integrable by Lemma 2.2. Subtracting their conditional expected logarithms proves (3.10). Including an independent mark η and then integrating it gives the stated column-noise extension.

For a continuous cylinder function g ,

$$Q^V(g) = \frac{Q(V^\theta(g \circ \text{sh}_d))}{Q(V^\theta)}.$$

The denominator has a positive uniform lower bound. These test functions determine the weak topology, proving continuity. Affinity is a property of the induced map on laws. The unconsumed

rows remain exchangeable. Successive tilts multiply the density by $U^\theta V^\theta$, so exchangeability of the fresh coordinate blocks gives commutation. Factoring the two normalizations gives (3.11). $\square$

For the vertex factor and the Poisson product of edge factors, define

$$(3.12) \quad P_\theta(\rho) = \psi_A^{(\theta)}(\rho) - \psi_B^{(\theta)}(\rho).$$

The sum over the new spin is already contained in A . Conditioning on the Poisson counts makes both factors positive cylinder functions. Lemma 2.1 then extends Lemma 3.2 to their averages by degree truncation. In particular, P_θ is continuous and affine on $\mathcal{C}$. We keep the whole Poisson product B in this functional. Reducing it to a multiple of a single-edge normalization would require single-edge stationarity as well as the cocycle.

3.4. Simultaneous stationarity and minimization. Let $\mathcal{S}_\theta \subseteq \mathcal{C}$ be the common fixed-point set of all positive continuous cylinder reweightings T_V . Define

$$(3.13) \quad p_\theta = \min_{\rho \in \mathcal{S}_\theta} P_\theta(\rho), \quad \mathcal{M}_\theta = \{\rho \in \mathcal{S}_\theta : P_\theta(\rho) = p_\theta\}.$$

Lemma 3.3. *The set $\mathcal{S}_\theta$ is nonempty, compact, and convex. The minimum in (3.13) exists, and $\mathcal{M}_\theta$ is a nonempty compact face of $\mathcal{S}_\theta$.*

If ρ is separately T_A - and T_B -stationary, a further commuting factor preserves both stationarities and each of the two expected normalizations $\psi_A^{(\theta)}$ and $\psi_B^{(\theta)}$.

Proof. Choose a countable dense family of positive continuous cylinder factors. The approximations may be chosen to preserve a positive lower bound for each factor being approximated. For the first m factors, with maps $T_1, \dots, T_m$, average an arbitrary $\rho_0 \in \mathcal{C}$ over a box of iterates:

$$\rho_{m,L} = \frac{1}{L^m} \sum_{0 \leq k_1, \dots, k_m < L} T_1^{k_1} \dots T_m^{k_m} \rho_0.$$

By commutation, applying any one of these m maps changes the average by at most $2/L$ in total variation norm. Compactness and continuity give a common fixed point for the first m maps. The resulting fixed-point sets are nested and compact, so their intersection is nonempty. Uniform approximation of factors extends stationarity to every positive continuous cylinder factor. The fixed-point set is compact and convex because the maps are continuous and affine. The same properties of P_θ give existence of the minimum, and affinity makes its minimizing set a face.

For the last assertion, compare the two orders in (3.11):

$$\psi_A^{(\theta)}(T_V \rho) - \psi_A^{(\theta)}(\rho) = \psi_V^{(\theta)}(T_A \rho) - \psi_V^{(\theta)}(\rho) = 0.$$

The same argument applies to B , and commutation preserves the two stationarities. The calculation also applies to any further reweighting for which commutation and the normalization cocycle have been established. $\square$

3.5. Comparison with the physical pressure. The law constructed in Proposition 3.1 is stationary under the ordinary tilts S_A, S_B . Introducing Poisson–Dirichlet weights changes the likelihood exponent from 1 to θ . The next estimate controls this change even after an arbitrary finite sequence of cylinder tilts.

Proposition 3.4. *Let $\mu \in \mathcal{C}$ be a law of a random directing measure G , stationary under S_A and S_B , and put*

$$P(\mu) = \mathbb{E} \log G(A) - \mathbb{E} \log G(B),$$

$$c_A = \mathbb{E} \operatorname{osc}_x \log A(x), \quad c_B = \mathbb{E} \operatorname{osc}_x \log B(x).$$

Assume these quantities are finite. Realize $\rho_0 = \text{Law}(G)$ with independent $\text{PD}(\theta, 0)$ weights. Every ρ in the closed convex hull of its finite cylinder-tilt orbit satisfies

$$(3.14) \quad -\frac{1-\theta}{\theta}c_A \leq P_\theta(\rho) - P(\mu) \leq \frac{1-\theta}{\theta}c_B.$$

Consequently

$$(3.15) \quad p_\theta \leq P(\mu) + \frac{1-\theta}{\theta}c_B.$$

For the present model, $c_A \leq 4\alpha c$ and $c_B \leq 2\alpha c$. In particular,

$$(3.16) \quad p_\theta \leq F + 2\alpha c \frac{1-\theta}{\theta}.$$

Proof. Consider a finite word of cylinder tilts. Before deleting its probe blocks, let L be the product of its likelihoods. Thus each deterministic factor V contributes V^θ , and a factor with an independent column-specific mark contributes $\mathbb{E}_\eta V^\theta$. The resulting directing measure, on the unconsumed coordinates, is

$$H = \frac{LG}{G(L)}.$$

As an ordinary reweighting, multiplication by L commutes with a fresh A tilt. The ordinary cocycle (3.2) and S_A -stationarity therefore give

$$(3.17) \quad \mathbb{E} \log H(A) = \mathbb{E} \log G(A).$$

The same identity holds for B . Shared external randomness in the word can first be conditioned on, so this argument also covers random factors.

For every probability measure H and bounded real function ℓ ,

$$(3.18) \quad -\frac{1-\theta}{\theta} \text{osc } \ell \leq \frac{1}{\theta} \log H(e^{\theta\ell}) - \log H(e^\ell) \leq 0.$$

The upper bound follows from concavity of $x \mapsto x^\theta$. For the lower bound,

$$H(e^{\theta\ell}) \geq e^{-(1-\theta) \sup \ell} H(e^\ell), \quad \log H(e^\ell) \geq \inf \ell.$$

Taking logarithms and combining these inequalities proves the claim. Apply (3.18) first to $\log A$ and then to $\log B$, average, and subtract. Equations (3.10) and (3.17) give (3.14) for each finite orbit word.

The error depends only on the oscillations of $\log A$ and $\log B$. It is independent of the length of the word and of the upper and lower bounds on L . Continuity and affinity therefore extend it to the closed convex orbit hull. That hull is invariant under every cylinder map. Applying the averaging argument of Lemma 3.3 within it gives a point of $\mathcal{S}_\theta$ that still satisfies (3.14). Minimization proves (3.15).

Finally, Lemma 2.1 gives

$$\text{osc } \log A \leq 2cD, \quad \text{osc } \log B \leq 2cK,$$

and hence the stated constants. Applying (3.15) to Proposition 3.1 proves (3.16). $\square$

We now keep $\theta < 1$ fixed. The remaining construction takes place inside $\mathcal{M}_\theta$. Its purpose is to obtain sufficient structure for finite-cascade approximation while preserving the value p_θ .

4. INTRINSIC FEATURES AND CLUSTER CENTRES

The variational minimum constructed in Section 3 is stationary under probes of finitely many spin coordinates. We shall enlarge the collection of coordinates that may be probed. The additional coordinates will record the centres of clusters of spin profiles and of features derived from them. Their role is to retain the information shared by particles in a cluster. To use these coordinates in a reweighting, we first need a measurable construction that is compatible with the directing measure and with the stationary operations.

Throughout this section, $0 < \theta < 1$ is fixed. We use the representation of Section 3: conditional on Q , the columns $X^a \in E$ are independent with law Q , and the decreasing weights (v_a) have law $\text{PD}(\theta, 0)$ independently. A feature is a bounded Borel map $\phi(Q, x) = (\phi(Q, x)_i)_{i \geq 1}$ that is equivariant under finite permutations of the site coordinates. Write $\phi^a(i) = \phi(Q, X^a)_i$. All feature families considered simultaneously are countable. Row exchangeability gives their empirical inner products

$$(4.1) \quad \langle \phi^a, \psi^b \rangle_{\text{emp}} = \lim_{L \rightarrow \infty} \frac{1}{L} \sum_{i=1}^L \phi^a(i) \psi^b(i).$$

These are conditional L^2 inner products in the directing representation of the exchangeable rows. They do not require independence of the rows conditional on their overlap array.

4.1. Intrinsic features and Gaussian probes.

Definition 4.1. A feature is *intrinsic* for a family of positive finite-row reweightings if its coordinates on the retained rows agree with their recomputed values after each corresponding equivalent positive change of directing measure. Its empirical inner products are also required to be unchanged by deleting finitely many rows. An admitted probe is a positive continuous function of finitely many admitted feature coordinates, with deterministic bounds $0 < u_- \leq U \leq u_+ < \infty$.

For a probe $U(Q, x)$ using d rows, its operation on the directing measure is

$$(4.2) \quad Q^U = (\text{sh}_d)_\# \frac{U(Q, \cdot)^\theta Q}{Q(U(Q, \cdot)^\theta)}, \quad T_U \rho = \text{Law}(Q^U).$$

Conditional on Q , the values $U(Q, X^a)$ are independent per-column marks. Thus Lemma 3.2 identifies this operation with multiplying the particle weights by $U(Q, X^a)$, normalizing, reordering, and deleting the probe rows.

The raw features $\phi^a(i) = \phi(X_i^a)$ are intrinsic. Bounded continuous coordinate functions of intrinsic features remain intrinsic. For a probe using a finite set of rows, intrinsicness concerns the rows remaining after that set has been deleted. We shall prove below that the same property holds for cluster centres.

The first consequence of probe stationarity is that empirical single-column moments cannot vary from one column to another. A second consequence is stationarity under Gaussian reweightings whose covariance is an empirical feature kernel.

Lemma 4.2. *Suppose that a law is stationary under all admitted probes of a countable family of bounded intrinsic features. For each such feature ϕ , the empirical mean*

$$d_\phi(a) = \lim_{L \rightarrow \infty} \frac{1}{L} \sum_{i=1}^L \phi^a(i)$$

is almost surely constant for Q -almost every column. If the stationary law is extreme in a convex class closed under conditioning on these common invariant moments, the common value is deterministic.

Let z be a bounded continuous function of finitely many admitted feature coordinates, and let I_j be disjoint row blocks of the required size. Put

$$(4.3) \quad K_{ab} = \lim_{L \rightarrow \infty} \frac{1}{L} \sum_{j=1}^L z^a(I_j) z^b(I_j).$$

If, conditional on the state, (g_a) is centred Gaussian with covariance K , then the full weighted marked law is stationary under

$$(4.4) \quad v_a \mapsto \frac{v_a e^{\lambda g_a}}{\sum_b v_b e^{\lambda g_b}}, \quad \lambda \in \mathbb{R},$$

followed by decreasing reordering. This also holds for finite sums and products of the covariance kernels so obtained. The covariance $K_0(a, b) = \mathbf{1}\{a = b\}$ may be added as separate label metadata.

Proof. The empirical means exist by row exchangeability. Apply stationarity to

$$U_L^a = \exp \left\{ \frac{\lambda}{L} \sum_{i=1}^L \phi^a(i) \right\},$$

using probe rows disjoint from the rows being observed. The factors have bounds independent of L . Convergence on finitely many columns and truncation of the independent Poisson–Dirichlet tail pass stationarity to the limiting factor $e^{\lambda d_\phi(a)}$. At the directing-measure level this factor has density $e^{\theta \lambda d}$, where $d = d_\phi$. For $\lambda > 0$,

$$(4.5) \quad \frac{Q(de^{\theta \lambda d})}{Q(e^{\theta \lambda d})} - Q(d) = \frac{\text{Cov}_Q(d, e^{\theta \lambda d})}{Q(e^{\theta \lambda d})} \geq 0.$$

Indeed, for independent d, d' with their Q -law, the covariance is one half of the expectation of $(d - d')(e^{\theta \lambda d} - e^{\theta \lambda d'})$. It vanishes precisely when d is Q -almost surely constant. Stationarity forces the nonnegative quantity in (4.5) to have expectation zero. The common mean is invariant under positive intrinsic tilts and finite row deletion. Conditioning on its invariant level sets preserves the stipulated class, so extremality makes it deterministic. Applying the same argument to admitted products of features gives the corresponding assertion for their actual self-overlap matrix.

For the Gaussian assertion, first rescale so that $|z| \leq 1$. Take common independent signs ε_j and use the factors

$$\exp(Y_a), \quad Y_a = \frac{\lambda}{\sqrt{L}} \sum_{j=1}^L \varepsilon_j z^a(I_j).$$

Conditional on the profiles, the finite-dimensional central limit theorem gives a centred Gaussian limit with covariance $\lambda^2 K$. We need a uniform tail estimate to normalize over all particles. Set $S_L = \sum_a v_a e^{Y_a}$. Jensen's inequality gives $S_L \geq \exp\{\sum_b v_b Y_b\}$. The signed increments in $Y_a - \sum_b v_b Y_b$ have interval length at most $4|\lambda|/\sqrt{L}$. The exponential moment bound for these increments therefore implies

$$(4.6) \quad \mathbb{E} \frac{\sum_{a>M} v_a e^{Y_a}}{S_L} \leq e^{8\lambda^2} \mathbb{E} \sum_{a>M} v_a.$$

The stated constant also permits an independent bounded diagonal completion. The right-hand side tends to zero uniformly in L , and the Gaussian limit satisfies the same bound.

Test first a finite cylinder of retained rows and origin labels, choosing all probe rows outside the observed row set. The conditional central limit theorem and (4.6) pass its law to the normalized Gaussian tilt. To pass the decreasing reordering, truncate to finitely many origin labels and then let the truncation increase. The output ranked weights have the fixed Poisson–Dirichlet law and have no ties. The resulting cylinder identities identify the entire weighted marked law. Equality of these laws then permits bounded measurable profile tests by a monotone-class argument.

Products of finite-row features on independent row blocks generate products of the kernels. Independent sign families generate sums, and linear combinations of feature coordinates give their joint quadratic forms. Independent column-specific Gaussian marks give K_0 directly by Poisson marking. This last operation concerns particle labels and leaves the physical feature collection unchanged. $\square$

In subsequent applications, Lemma 4.2 is applied to the feature family already admitted at the current stage. Gaussian stationarity is therefore available for its full marked law before any new centre is introduced.

4.2. Centres of sampled cluster hulls. Fix an admitted scalar feature ϕ , normalized so that $|\phi| \leq 1$, and a measurable partition kernel built from the admitted empirical Gram array. Let q be a fixed cut for which

$$C_q(x) = \{y : K(x, y) \geq q\}$$

defines an equivalence class for almost every independently sampled reference. Conditional on (Q, x) , draw independent helpers $Y^1, Y^2, \dots$ with law Q . Retain those in $C_q(x)$ and include the reference x . The empirical Gram matrix of their feature profiles realizes this countable collection in a common Hilbert space. Let H_M be the convex hull using the reference and the first M retained-or-rejected helper draws, and set

$$H = \overline{\bigcup_{M \geq 1} H_M}.$$

The *cluster centre* $c_q\phi(Q, x)$ is the minimum-norm point of this sampled hull, with coordinate version constructed below.

The sampled interpretation also specifies the endpoints of the construction. If $Q(C_q(x)) = 0$, no helper is retained almost surely and the centre is $\phi(Q, x)$. At the separate label-identity level, the class is the single sampled label, whose centre is again its own feature. This convention applies even when two labels carry the same profile.

Lemma 4.3. *For almost every sampled reference, the sampled hull has a unique minimum-norm point. Its value is independent of the helper sequence and agrees with the centre formed from the actual countable particle set. Within a positive- Q -mass class it is independent of the reference, for almost every reference pair in that class.*

There is one Borel, row-equivariant coordinate map $c_q\phi(Q, x)$ realizing these centres under every admissible coordinate-exchangeable law. If the old profiles and partition kernel are carried unchanged in the empirical Hilbert space, an equivalent positive change of directing measure preserves the centre in that space. For an intrinsic finite-row probe, the canonical centre coordinates agree on every retained row. Consequently the centre is an intrinsic feature.

Proof. We give the geometric construction first and then choose its coordinate version.

Projection and helper independence. Let b_M minimize the norm on H_M . A norm-minimizing sequence in the closed convex hull H is Cauchy by the parallelogram identity. Its limit belongs to H , and strict convexity makes the limit b unique. Projection of zero onto H gives $\langle b_M - b, b \rangle_{\text{emp}} \geq 0$. Since the finite hulls increase to a dense subset of H ,

$$(4.7) \quad \|b_M - b\|_{\text{emp}}^2 \leq \|b_M\|_{\text{emp}}^2 - \|b\|_{\text{emp}}^2 \longrightarrow 0.$$

Conditional on (Q, x) , the limiting minimum norm is invariant under every finite permutation of the independent helpers. The permutation zero-one law makes it deterministic. Two independent helper sequences and their interleaved union have the same conditional law for this limiting norm. In their common Gram realization, the two separate centres lie in the union hull and attain its minimum norm. Uniqueness identifies them. The actual unweighted columns other than a fixed reference label form another independent Q -sequence conditional on (Q, x) . Comparing this sequence with

independent helpers proves the asserted identification with the actual particle hull. Countability makes the identification simultaneous for all reference labels under consideration.

For reference independence in a positive-mass class, define

$$(4.8) \quad T_x(dy) = \frac{\mathbf{1}\{y \in C_q(x)\}Q(dy)}{Q(C_q(x))}.$$

This measurable measure-valued identifier is constant on a class, and different classes give mutually singular measures. Conditional on Q and the identifier T , a sampled reference has law T . The accepted helpers form an independent iid T -sequence. Prepending the reference leaves its limiting minimum norm unchanged. Two references and their helper sequences can therefore be compared in a common union hull, where uniqueness again identifies the projections. A zero-mass class is a singleton in the sampled exchangeable partition and is covered by the convention above.

From empirical equality to coordinate equality. If bounded coupled profiles B and $\tilde{B}$ have a jointly exchangeable row law, then

$$(4.9) \quad \mathbb{E}|B_1 - \tilde{B}_1|^2 = \lim_{L \rightarrow \infty} \mathbb{E} \frac{1}{L} \sum_{i=1}^L |B_i - \tilde{B}_i|^2.$$

Thus zero empirical Hilbert distance implies equality at each fixed coordinate. Bounded convergence justifies the limit, and countability gives equality simultaneously at all coordinates used here. We use this observation on retained rows after a finite-row probe has been deleted.

A universal Borel version. Empirical Gram limits, finite positive-semidefiniteness tests, and the minimum of a quadratic form over a finite simplex are Borel. Assign a fixed fallback value if an empirical limit is undefined or a positive-semidefiniteness test fails. These exceptional inputs have zero probability under the admissible laws.

Write $d_M(Q, x, Y)$ for the squared minimum norm of the finite selected hull and put

$$(4.10) \quad m(Q, x) = \inf_{M \geq 1} \operatorname{ess\,inf}_{Y \sim Q^{\otimes M}} d_M(Q, x, Y).$$

The strict sublevel sets of m are countable unions of the measurable conditions $Q^{\otimes M}\{d_M < t\} > 0$, with rational t . Hence m is Borel. Independent disjoint helper blocks show that the limiting minimum squared norm is $m(Q, x)$ almost surely. The lower bound follows from the essential infima. For the upper bound, every strict upper bound on m has a finite helper event of positive probability, and one of the disjoint blocks eventually realizes that event.

For $\epsilon > 0$, choose the least M for which

$$Q^{\otimes M}\{d_M \leq m + \epsilon\} > \frac{1}{2}.$$

This choice is measurable and finite. On its success event, let H_M^{Gram} be the Gram matrix of the selected profiles and minimize

$$\lambda^\top H_M^{\text{Gram}} \lambda + \epsilon \|\lambda\|_2^2$$

over their coefficient simplex. The minimizer is unique and measurable. Its coordinate-valued convex combination B_ϵ satisfies $\|B_\epsilon\|_{\text{emp}}^2 \leq m + 2\epsilon$, so projection gives

$$(4.11) \quad \|B_\epsilon - b\|_{\text{emp}}^2 \leq 2\epsilon.$$

Define $c_{\epsilon,i}(Q, x)$ by averaging $B_{\epsilon,i}$ over the successful helper tuples. These are Borel integrals, and their denominator is greater than $1/2$. Set them to zero on exceptional inputs where the choice fails.

These coordinate averages also represent points of the common closed hull. To verify this using only countably many profiles, draw independent successful helper tuples. Conditional on (Q, x) , the coordinate averages of the resulting B_ϵ converge to c_ϵ . The expected empirical squared error of the average of L tuples is at most $1/L$, because the coordinates are bounded by one. Along a summable subsequence the averages converge in the empirical Hilbert norm as well. Every average belongs to the hull and, by (4.11), lies within $\sqrt{2\epsilon}$ of its centre. The same is therefore true of c_ϵ .

Comparison of two accuracies, coordinatewise Jensen’s inequality, and (4.9) give, under every admissible law,

$$(4.12) \quad \mathbb{E}|c_{\epsilon,i}(Q, X) - c_{\epsilon',i}(Q, X)|^2 \leq 4(\epsilon + \epsilon').$$

Choose $\epsilon_k = 2^{-4k}$. The expected sum of the absolute successive differences is finite, so this full sequence converges almost surely at each coordinate. Define

$$(4.13) \quad c_Q \phi(Q, x)_i = \lim_{k \rightarrow \infty} c_{\epsilon_k, i}(Q, x)$$

where the limit exists, and use zero otherwise. This is one Borel, row-equivariant map. Bounded coordinate convergence and row exchangeability identify its empirical Hilbert vector with b . The construction depends on (Q, x) and the admitted features, so the same map is available on every admissible image law. Applying (4.9) to the preceding union constructions also gives their helper and reference independence at every fixed coordinate.

Equivalent changes of directing measure. Suppose that $\tilde{Q}$ is equivalent to Q and the old profiles and partition kernel are carried unchanged in the empirical Hilbert space. A strict upper bound on the limiting minimum norm has a finite witness: finitely many helpers, their class memberships, and a convex combination with rational coefficients. Such a finite event has positive probability under Q exactly when it does under $\tilde{Q}$. The conditional limiting minimum norms therefore agree. The same argument applies in a union of helper samples from the two measures, so uniqueness of projection identifies their Hilbert centres. This applies within each positive-mass class through the restrictions in (4.8), and equivalence preserves the zero-mass convention.

For a finite-row intrinsic probe, the coupled construction is exchangeable in the retained rows. Equation (4.9) therefore identifies the canonical centre coordinates on those rows. Deleting finitely many rows leaves all empirical products unchanged. These are the two requirements of intrinsicness. $\square$

The preceding construction establishes measurability and compatibility of centres with positive reweightings. The uniform approximation needed to average the new probe maps on a compact class is proved in Lemma 5.3, after the relevant Gram law has been fixed.

4.3. Gaussian preservation of canonical centres. We now compare a newly adjoined centre with its value after an old Gaussian operation. Fix an old empirical covariance K with deterministic diagonal $K_{aa} = d$. Let $\mathcal{G}$ multiply the weights by e^{g_a} , normalize, and reorder, where (g_a) is conditionally centred Gaussian with covariance K . A finite amplitude can be absorbed in K . The operation deletes no rows.

Proposition 4.4. *Assume that $\mathcal{G}\rho = \rho$ as laws of the full old weighted marked array. Adjoin finitely many sampled-hull centres of old features at cuts of old partition kernels. Under $\mathcal{G}$, the centres carried with each particle agree almost surely with the canonical centres recomputed from the output directing measure. After matching particle labels, the equality holds jointly at every fixed site coordinate.*

Proof. The normalization is finite and positive, since

$$\mathbb{E}_g \sum_a v_a e^{g_a} = e^{d/2} < \infty.$$

Every multiplier is strictly positive, so every original particle has positive output weight. A positive summable sequence has only finitely many entries above any fixed positive threshold. Sorting therefore permutes the same countable particle set. Ties have probability zero: if the difference of two Gaussian increments has positive variance this follows from its continuous law, and otherwise the increments agree while the original Poisson–Dirichlet weights are distinct.

The carried old profiles, their empirical Gram kernels, and their partition classes are unchanged after the labels have been transported. Thus each hull formed from the actual countable particle

set is the same subset of the common empirical Hilbert space before and after the operation. Its minimum-norm point is unchanged.

To identify this point with a canonical output centre, use the assumed stationarity of the full old law. The output unweighted ranked columns have an iid directing representation. Its directing measure is recovered almost surely in the weak topology by

$$(4.14) \quad Q^{\mathcal{G}} = \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{a=1}^M \delta_{X^{\mathcal{G}}, a}.$$

This is a Borel reconstruction that commutes with finite row permutations. Lemma 4.3, applied to this output representation, identifies its canonical centre with the centre of the output actual particle hull. That hull is the input hull, so the canonical centres agree in the empirical Hilbert space.

Finally, K is invariant under finite permutations of site rows. The coupling of the input state, Gaussian field, reordering, and output state is consequently row exchangeable. After matching labels, (4.9) promotes the Hilbert equality to coordinate equality. The construction is countable, giving the asserted joint conclusion. $\square$

The measure $Q^{\mathcal{G}}$ is reconstructed from the new ranking. The object preserved in this argument is the actual countable particle hull. Gaussian stationarity supplies the output directing representation that identifies this hull with the canonical output centre.

4.4. Commutation with a new centre probe. Let U be a new admitted factor built from old features and the centres just constructed. As before, T_U reweights particle a by U_a , normalizes, reorders, and deletes its finite probe-row set.

Proposition 4.5. *Under the assumptions of Proposition 4.4, the two orders $\mathcal{G}T_U$ and $T_U\mathcal{G}$ have the same law on the raw array and the canonically marked old features and new centres, with the probe rows deleted. In particular,*

$$(4.15) \quad \mathcal{G}(T_U\rho) = T_U\rho.$$

Proof. Sample the old Gaussian field independently of the auxiliary randomness of U , conditional on the input state.

Apply T_U first. Intrinsicness preserves the empirical covariance K after deleting the probe rows. Transport the old field through the U -reordering. Conditional on the input and the U randomness, the transported field is centred Gaussian with covariance equal to the kernel of the output state. This covariance is measurable from that state, so conditioning only on the output state gives the same Gaussian law. Hence the transported field realizes the second operation in $\mathcal{G}T_U\rho$.

Apply $\mathcal{G}$ first. The assumption $\mathcal{G}\rho = \rho$ concerns the old stationary law at which Proposition 4.4 applies. That proposition identifies the recomputed centres used by U with their carried coordinates. The factor evaluated on each old particle is therefore its original U_a .

In either order, the final weights before decreasing reordering are

$$(4.16) \quad \frac{v_a U_a e^{g_a}}{\sum_b v_b U_b e^{g_b}},$$

and the same probe rows are deleted. Thus the final raw arrays and carried marks agree after matching labels.

The Gaussian-first order has the canonical directing representation: old Gaussian stationarity supplies it after $\mathcal{G}$, and Poisson marking supplies it after T_U . Equality of the final ranked arrays and the reconstruction (4.14) identify the directing measure in the other order as well. The universal maps of Lemma 4.3 then identify its canonical centre marks. We have proved equality of the canonically marked laws $\mathcal{G}T_U\rho = T_U\mathcal{G}\rho$. Substitution of $\mathcal{G}\rho = \rho$ gives (4.15). $\square$

The new probe therefore preserves every old Gaussian stationarity for which the hypotheses above hold. The next section uses this commutation to preserve the old marked identities and their constants, and then to construct stationary laws for the enlarged feature family.

5. MARKED IDENTITIES AND STATIONARY ENRICHMENT

We now construct stationary laws that retain the additional features introduced in Section 4. The construction takes place inside the minimizing class $\mathcal{M}_\theta$ of Section 3. At each stage we first fix the overlap data of the admitted features. We can then add their cluster centres, average the new probe maps, and fix the enlarged overlap data. The order matters because continuity of a centre operation will be proved uniformly on a class with fixed operand and partition Gram laws.

Two facts make this induction possible. Extremality turns a stationary Gaussian normalization velocity into a deterministic constant and thereby gives marked Ghirlanda–Guerra identities. Once these identities are available, an exact second-moment estimate preserves both the identities and their constants under subsequent intrinsic tilts. We establish these facts before proving the continuity and compactness statements needed for enrichment.

5.1. Marked identities at an extreme stationary law. Let S denote the weighted marked state. An admitted covariance K is a positive semidefinite kernel obtained from the finite-row feature probes of Lemma 4.2, including their finite sums and products and the separate label-identity covariance. Throughout this subsection its actual diagonal is the deterministic constant

$$K_{aa} = d.$$

Given S , let $(g_a)_a$ be centred Gaussian with covariance K , and define

$$(5.1) \quad F_{T,\lambda}(S) = \log \sum_a v_a e^{\lambda \sqrt{T} g_a}, \quad T > 0, \quad \lambda \in \mathbb{R}.$$

The bracket $\langle \cdot \rangle_{T,\lambda}$ uses the corresponding reweighted Gibbs measure. A sampled-mark test is a bounded measurable function of the marks and overlaps of finitely many Gibbs replicas. In particular, it does not test the changing numerical weights themselves.

Lemma 5.1 (Marked identities). *Let $\mathcal{D}$ be a compact convex class of laws stationary under the admitted intrinsic finite-row probes. Suppose that conditioning on an event measurable from Q and invariant under these probes and finite row permutations preserves $\mathcal{D}$. Let ρ be an extreme point of $\mathcal{D}$, and suppose the admitted covariance K has deterministic diagonal d . Then, with $m = \mathbb{E}_\rho \langle K_{12} \rangle$, every sampled-mark test f of n replicas satisfies*

$$(5.2) \quad n \mathbb{E}_\rho \langle f K_{1,n+1} \rangle = m \mathbb{E}_\rho \langle f \rangle + \sum_{\ell=2}^n \mathbb{E}_\rho \langle f K_{1\ell} \rangle.$$

For the polynomial covariance probes, these are the generalized matrix Ghirlanda–Guerra identities with the same sampled-mark tests.

Proof. We first justify differentiation of the normalization over its countably many particles. Put $\bar{g} = \sum_a v_a g_a$. Jensen’s inequality gives a denominator at least $e^{\lambda \sqrt{T} \bar{g}}$, and $\text{Var}_g(g_a - \bar{g}) \leq 4d$. Consequently, for every integer $k \geq 1$,

$$(5.3) \quad \begin{aligned} \mathbb{E}_g \langle |g|^k \rangle_{T,\lambda} &\leq \sum_a v_a \mathbb{E}_g [|g_a|^k e^{\lambda \sqrt{T} (g_a - \bar{g})}] \\ &\leq (\mathbb{E} |N(0, d)|^{2k})^{1/2} e^{4\lambda^2 T d}. \end{aligned}$$

The bound also holds for every normalized finite truncation of the weights. Applying it with higher moments gives uniform integrability. The corresponding absolute-exponential estimate on compact amplitude intervals justifies differentiation under expectation. Thus finite-particle Gaussian integration by parts and the first two normalization derivatives pass to the full weighted array.

We next identify the limiting normalization velocity. Realize the Gaussian fields at integer times by independent increments, each with covariance K . Full marked stationarity, supplied by Lemma 4.2, makes the successive logarithmic normalization increments stationary. The conditional variance of one increment is at most $\lambda^2 d$ by Gaussian Poincaré, while its conditional mean lies between 0 and $\lambda^2 d/2$ by Jensen's inequality and Gaussian exponential moments. Its second moment is therefore finite. The L^2 mean ergodic theorem gives a limit for $F_{T,\lambda}/T$ along integer times. Moreover,

$$(5.4) \quad \text{Var}_g(F_{T,\lambda} \mid S) \leq \lambda^2 T d.$$

After division by T^2 , this shows that the limit is a measurable function $c_{K,\lambda}(S)$ of the initial state.

This velocity is unchanged by bounded positive changes in the ratios of the weights. It is also unchanged by deleting finitely many particles. To see the latter assertion, normalize the remaining weights and lower-bound their log normalization by their weighted Gaussian mean. Each deleted term, divided by this lower bound in exponential form, has a Gaussian logarithm with variance at most $4\lambda^2 T d$, together with a fixed logarithmic weight ratio. A finite log-sum bound yields

$$(5.5) \quad |\mathbb{E}_g F_{T,\lambda}(S) - \mathbb{E}_g F_{T,\lambda}(S \text{ with finitely many particles deleted})| \leq C_S(1 + \sqrt{T}).$$

The constant may depend on the original weights and the deleted set. Changing finitely many columns has the same negligible effect, by deleting those columns in both configurations.

For completeness, we make the resulting conditional tail argument explicit. Represent the Poisson–Dirichlet weights as

$$(5.6) \quad v_a = \frac{\Gamma_a^{-1/\theta}}{\sum_b \Gamma_b^{-1/\theta}}, \quad \Gamma_a = e_1 + \cdots + e_a,$$

where the e_a are independent unit exponentials, independent of the columns, which are conditionally iid given Q . Resampling finitely many e_a changes all weight ratios by bounded positive factors. Conditional on Q , resampling finitely many columns leaves every remaining canonical feature equal to the same function of Q and its unchanged column. Its restricted Gaussian kernel is therefore unchanged.

Set $f_T(S) = \mathbb{E}_g F_{T,\lambda}(S)$. The L^2 convergence above and (5.4) give $f_T/T \rightarrow c_{K,\lambda}(S)$ in L^2 . Choose one deterministic subsequence $T_j \uparrow \infty$ along which the squared errors are summable. By Fubini, for almost every Q the convergence set has full conditional product measure in $(e_a, X^a)_{a \geq 1}$. Finite resampling preserves this product measure, so both configurations lie in that convergence set almost surely. The bounded ratio comparison and (5.5), divided by T_j , identify their limits pointwise. Thus the velocity is measurable with respect to every completed tail sigma field of this conditionally independent sequence. The conditional zero-one law gives

$$c_{K,\lambda}(S) = c_{K,\lambda}(Q) \quad \text{almost surely.}$$

This argument only uses pointwise comparison constants in (5.5).

Let U be a defining bounded positive intrinsic factor. Couple the future Gaussian field before and after its tilt. Intrinsicness leaves the kernel on the retained rows unchanged, and normalization gives

$$(5.7) \quad |F_{T,\lambda}(S^U) - F_{T,\lambda}(S)| \leq \text{osc}_a \log U_a.$$

Hence $c_{K,\lambda}(Q^U) = c_{K,\lambda}(Q)$. A finite row permutation leaves the empirical covariance and normalization unchanged as well. The level-set events of this velocity therefore satisfy the conditioning assumption of the lemma. Extremality makes the velocity deterministic.

Gaussian integration by parts and stationarity give

$$(5.8) \quad \mathbb{E}_\rho F_{T,\lambda} = \frac{T\lambda^2}{2}(d - m).$$

The deterministic velocity is consequently $\lambda^2(d-m)/2$. Apply the preceding argument on a countable dense set of amplitudes. Convex secants and their L^2 bounds imply convergence in L^1 of the normalized derivative,

$$\left\langle \frac{g_1}{\sqrt{T}} \right\rangle_{T,\lambda} \longrightarrow \lambda(d-m).$$

Differentiating (5.8) also gives

$$(5.9) \quad \mathbb{E}_\rho \operatorname{Var}_{\langle \cdot \rangle_{T,\lambda}} \left(\frac{g}{\sqrt{T}} \right) = \frac{d-m}{T}.$$

It is enough to use amplitudes at which these derivatives hold. Thus $g_1/\sqrt{T} \rightarrow \lambda(d-m)$ in joint quenched and Gibbs L^1 .

During Gaussian differentiation, evaluate f on the carried initial marks. Their derivatives are zero. Proposition 4.4 identifies these marks with the canonical output marks, so full marked stationarity applies to the resulting expression. Integration by parts yields

$$(5.10) \quad \mathbb{E}_\rho \left\langle f \frac{g_1}{\sqrt{T}} \right\rangle_{T,\lambda} = \lambda \mathbb{E}_\rho \left\langle f \left(d + \sum_{\ell=2}^n K_{1\ell} - nK_{1,n+1} \right) \right\rangle_{T,\lambda}.$$

Stationarity removes the tilt on the right. Passing to the energy limit on the left, subtracting $\lambda(d-m)\mathbb{E}_\rho \langle f \rangle$, and dividing by a nonzero λ proves (5.2).

For a finite feature vector $(\phi_1, \dots, \phi_s)$, write $R_{ab}^{jk} = \langle \phi_j^a, \phi_k^b \rangle$ for its empirical overlap block. The row-block probe

$$z^a(i_1, \dots, i_p) = \sum_{j=1}^s c_j \prod_{b=1}^p \phi_j^a(i_b)$$

has covariance $c^\top R_{ab}^{op} c$. Products of these forms are generated on disjoint row blocks. Applying (5.2) to these covariances gives the asserted generalized identities. $\square$

5.2. Preserving the old marked law. The next estimate will allow us to retain the constants in (5.2) while enlarging the permitted tests. Its hypotheses involve only an already established stationary marked law.

Lemma 5.2 (Quantitative drift and preservation). *Suppose ρ is Gaussian stationary for K , $K_{aa} = d$ is deterministic, and (5.2) holds with $m = \mathbb{E}_\rho \langle K_{12} \rangle$. Then*

$$(5.11) \quad \mathbb{E}_\rho \left\langle \left(\frac{g_1}{\sqrt{T}} - \lambda(d-m) \right)^2 \right\rangle_{T,\lambda} = \frac{d}{T},$$

$$(5.12) \quad \left\| \frac{F_{T,\lambda}}{T} - \frac{\lambda^2}{2}(d-m) \right\|_{L^2} \leq |\lambda| \sqrt{\frac{d}{T}}.$$

Let U be bounded, positive and intrinsic, and suppose its exact map commutes with the old Gaussian maps. The image law $T_{U\rho}$ satisfies the same identities with the same m , including tests of any newly admitted canonical centres carried by the Gaussian reweighting. Applying this conclusion to every old polynomial covariance preserves their joint pair law. The same preservation holds after conditioning on an invariant event of positive probability, provided the conditional law retains the defining finite-row stationarities.

Proof. The integrability argument (5.3) justifies two Gaussian integrations by parts. They give

$$\begin{aligned}\mathbb{E}_\rho \left\langle \frac{g_1}{\sqrt{T}} \right\rangle_{T,\lambda} &= \lambda(d-m), \\ \mathbb{E}_\rho \left\langle \frac{g_1^2}{T} \right\rangle_{T,\lambda} &= \frac{d}{T} + \lambda^2 \mathbb{E}_\rho \langle (d-K_{12})(d+K_{12}-2K_{13}) \rangle.\end{aligned}$$

The identity with $n=2$ and $f=K_{12}$ reads

$$2\mathbb{E}_\rho \langle K_{12}K_{13} \rangle = m^2 + \mathbb{E}_\rho \langle K_{12}^2 \rangle.$$

It reduces the last expectation to $(d-m)^2$, which proves (5.11). Since $\partial_\lambda(F_{T,\lambda}/T) = \langle g_1/\sqrt{T} \rangle_{T,\lambda}$, Jensen's inequality bounds the L^2 norm of the centred derivative by $\sqrt{d/T}$. Integrating from 0 to λ and applying Minkowski gives (5.12).

Write $u_- \leq U_a \leq u_+$ and $\rho^U = T_U \rho$. The coupling in (5.7) gives the more explicit estimate

$$(5.13) \quad \left\| \frac{F_{T,\lambda}(S^U)}{T} - \frac{\lambda^2}{2}(d-m) \right\|_{L^2} \leq |\lambda| \sqrt{\frac{d}{T}} + \frac{\log(u_+/u_-)}{T}.$$

Indeed, if $Z = \sum_a v_a e^{\lambda\sqrt{T}g_a}$ and $Z_U = \sum_a v_a U_a e^{\lambda\sqrt{T}g_a}$, then

$$F_{T,\lambda}(S^U) - F_{T,\lambda}(S) = \log(Z_U/Z) - \log \sum_a v_a U_a,$$

and both arguments of the logarithms belong to $[u_-, u_+]$.

Commutation, as established in Proposition 4.5 for the centre maps, makes ρ^U Gaussian stationary. Setting $m_U = \mathbb{E}_{\rho^U} \langle K_{12} \rangle$, integration by parts gives

$$\mathbb{E}_{\rho^U} F_{T,\lambda} = \frac{T\lambda^2}{2}(d-m_U).$$

Comparison with (5.13) at a nonzero amplitude shows $m_U = m$. Convex secants and the thermal-variance calculation (5.9) now give the same joint energy limit under ρ^U . For a test f using the new centre coordinates, carry the initial marks during differentiation and use Propositions 4.4 and 4.5 to identify the canonical output marks. Formula (5.10) then proves (5.2) under ρ^U , with this enlarged f and the old constant m .

For the vector application, the generalized identities and finite-dimensional synchronization make the old overlap blocks symmetric almost surely. Equivalent positive tilts and finite row deletion preserve this property. Rational quadratic forms separate symmetric matrices, and mixed polynomial moments of these forms determine their joint law on its compact range. Applying the preceding argument to the old polynomial covariances therefore preserves the full old joint pair law.

Finally, let H be an invariant event of probability $p > 0$ measurable from Q . By assumption its conditional law retains the finite-row stationarities, and Lemma 4.2 supplies Gaussian stationarity there. Restrict the Gaussian coupling to H . The squared error in (5.12) increases by at most p^{-1} , so its deterministic limiting velocity is unchanged. The same expectation comparison and marked integration by parts recover the old constants and identities under this conditional law. $\square$

5.3. Uniform centre approximation and continuity. We use a precise approximation property during the induction. For every finite list of admitted feature coordinates, require a joint mean-square approximation, uniform over the current class $\mathcal{D}$, by bounded continuous functions of a finite raw profile subarray and finitely many independent Q -helper columns. The classification rows used by an approximation are disjoint from its evaluation rows and all retained output rows. Ordinary features have this property, and bounded continuous coordinate functions preserve it by uniform continuity on compact sets.

Lemma 5.3 (Uniform approximation and Feller centre maps). *Let $\mathcal{D} \subseteq \mathcal{C}$ have a fixed full joint unweighted Gram-array law for a finite family of admitted features and an old partition kernel. Suppose the old features have the uniform approximation property above. Choose finitely many of their actual centres at fixed included thresholds of that partition. Then these centres have the same uniform approximation property.*

If U is a bounded positive continuous function of finitely many old and new feature coordinates, with $0 < u_- \leq U \leq u_+$, its exact intrinsic probe is a continuous affine map from $\mathcal{D}$ to $\mathcal{C}$. It preserves the independent Poisson–Dirichlet construction and commutes with the other exact intrinsic fresh-row maps.

Proof. Rescale the finitely many scalar features to be bounded by one. Fix a reference column x , a threshold q , and a feature ϕ . Let b_M be the minimum-norm point of the hull of its reference vector and those among the first M independent helper draws that belong to $C_q(x)$, and let $b = c_q \phi^x$ be the actual sampled-hull centre from Lemma 4.3. The projection inequality gives

$$(5.14) \quad \|b_M - b\|^2 \leq \|b_M\|^2 - \|b\|^2.$$

The nonnegative right side is a function of the joint Gram array of the reference, helpers and partition kernel. Its law is fixed on $\mathcal{D}$, and it converges boundedly to zero. Hence

$$(5.15) \quad \sup_{\rho \in \mathcal{D}} \mathbb{E}_\rho[\|b_M\|^2 - \|b\|^2] \rightarrow 0.$$

The identity level has a singleton sampled class and centre ϕ^x , so it already satisfies the required assertion.

For a fixed selected list with Gram matrix H , minimize

$$(5.16) \quad a^\top H a + \varepsilon \|a\|_2^2 \quad \text{over the coefficient simplex.}$$

Strict convexity and compactness make its unique coefficients continuous in H . The resulting vector $b_{M,\varepsilon}$ satisfies

$$(5.17) \quad \|b_{M,\varepsilon} - b\|^2 \leq \|b_M\|^2 - \|b\|^2 + \varepsilon.$$

Changing every Gram entry by at most η changes every simplex quadratic objective by at most η . Comparing the two minimizing objectives adds at most 2η to this bound.

Estimate the Gram entries with L fresh classification rows. Conditional row independence gives

$$(5.18) \quad \mathbb{E} \left| \frac{1}{L} \sum_{i=1}^L \phi^a(i) \phi^b(i) - \langle \phi^a, \phi^b \rangle \right|^2 \leq \frac{1}{L}.$$

The same estimate, after bounded rescaling, applies to cross-feature entries. To approximate membership at an included threshold, take a continuous $t_j \in [0, 1]$ which is zero when the estimated partition overlap between x and helper j is at most $q - 3\eta$ and one when it is at least $q - 2\eta$. When the estimation error is at most η , this agrees with the exact membership except possibly when the true overlap lies in $[q - 4\eta, q)$. Its exceptional probability is bounded by

$$(5.19) \quad \mathbb{P}(q - 4\eta \leq R_{0j} < q) + \frac{C}{L\eta^2}.$$

The first term tends to zero even when q is an atom. Its value is uniform over $\mathcal{D}$ because the pair law is fixed.

Keep a fixed list of surrogate vectors

$$z_0 = \phi^x, \quad z_j = t_j \phi^{Y^j} + (1 - t_j) \phi^x, \quad 1 \leq j \leq M.$$

If the memberships are exact, excluded helpers are duplicate reference vectors and this is the selected hull. Let $A(t)$ be the row-stochastic matrix implementing the combinations. Build the empirical Gram matrix $\hat{H}$ from one common list of feature evaluations and use $A(t) \hat{H} A(t)^\top$ in (5.16). This

remains positive semidefinite. Its entries, minimizing coefficients and evaluated convex combination are continuous functions of the fixed finite array once the old coordinates have been approximated.

On the event of correct classification, (5.17) and the Gram-error estimate apply. Multiplication by the row-stochastic matrices preserves an entrywise error bound, and duplicating the reference does not change the unregularized hull minimum. On the exceptional event, both vectors have coordinates bounded by one, giving squared coordinate error at most four. A finite union bound controls all classifications. An independent evaluation row converts the empirical Hilbert bound into the expected coordinate bound by exchangeability outside the classification block. This also treats any fixed finite list of evaluation rows.

The choices are made in the following order. First choose M by (5.15). Then choose ε and the membership band η . Next choose L so that the finite union bounds and $L\eta^2$ are sufficiently large. Finally approximate the finitely many old-feature evaluations finely enough for uniform continuity of these fixed finite operations. The classification rows are deleted. This proves the new uniform approximation property in terms of raw profile arrays and independent helpers.

To prove continuity of the induced law, use one shared auxiliary helper list $Y = (Y^1, \dots, Y^M)$, independent of the weighted columns conditional on Q , and write $U^{\text{app}}(Q, x, Y)$ for the approximating factor, truncated to $[u_-, u_+]$. Conditional on (Q, Y) this remains a per-column factor. Delete the classification rows in both the exact and approximate operations. Deleting these additional unused rows leaves the exact output law unchanged by exchangeability. Poisson marking, Lemma 3.2, gives its output directing density proportional to $(U^{\text{app}})^\theta$. For $|g| \leq 1$,

$$(5.20) \quad \left| \frac{Q(U^\theta g)}{Q(U^\theta)} - \frac{Q((U^{\text{app}})^\theta g)}{Q((U^{\text{app}})^\theta)} \right| \leq \frac{2\theta}{u_-} Q|U - U^{\text{app}}|.$$

Indeed $t \mapsto t^\theta$ is $\theta u_-^{\theta-1}$ -Lipschitz on this interval and both denominators are at least u_-^θ . For a test of k output columns, tensorization multiplies the bound by at most k .

Each approximate map is continuous on laws of Q : it consists of finitely many continuous cylinder integrals, divisions by denominators bounded away from zero, and integration against $Q^{\otimes M}$. Uniform approximation in (5.20) proves that the exact map is Feller. It is affine because it acts as a Markov kernel on the law of Q . Poisson marking also preserves the independent Poisson–Dirichlet representation. Finally, intrinsicness identifies the centres on the untouched rows after each positive tilt. Interchanging the two finite probe blocks before deleting them proves commutation of the exact maps. $\square$

5.4. The finite enrichment step. We recall how finite-dimensional synchronization enters the construction. For a finite bounded feature vector with a deterministic actual self-overlap matrix, the generalized identities generated in Lemma 5.1 are those in [7, Theorem 4, equation (88)]. The theorem identifies its overlap matrix as a deterministic function of its scalar trace. The scalar identities make that trace ultrametric by [3]. Its pair law determines the full Gram-array law: a new replica joins an old threshold class C at level t with conditional probability $[\mu([t, \infty)) + |C| - 1]/n$. The recursive determination is given in Lemma 6.2. These finite-dimensional facts will be used with each old family before its centres are added.

Theorem 5.4 (Finite enrichment). *Let $\mathcal{D}$ be a nonempty compact convex subset of $\mathcal{M}_\theta$ defined by the following fixed data: stationarity under its admitted intrinsic probes, deterministic actual feature self-overlaps, generalized marked Ghirlanda–Guerra identities, and a fixed joint pair law for the old feature kernels. Suppose the old features have the uniform finite-array approximation property of Lemma 5.3, and the finite synchronization statement above applies.*

Choose finitely many admitted scalar features and their centres at fixed thresholds of an old finite master trace. Each centre operand and partition kernel belongs to the old family with fixed complete Gram law. Adjoin these centres and finitely many bounded continuous coordinate functions or coordinatewise products of the resulting list. Then $\mathcal{D}$ contains a nonempty compact convex subclass

on which all bounded positive continuous finite-row probes of the new features are stationary, the enlarged actual self-overlaps are deterministic, and the enlarged generalized marked identities and joint pair law are fixed. Every law in this subclass has value p_θ , and the new features have the same uniform approximation property.

A centre of a newly formed coordinate function is admitted at a subsequent stage, after that function's Gram data have been fixed.

Proof. Synchronization and scalar sampling fix the complete old weighted Gram-array law. They also fix its unweighted helper version. To verify this, retain the label-identity metadata and condition k Gibbs replicas to have distinct labels. For $\text{PD}(\theta, 0)$ weights this event has probability $\theta^{k-1} > 0$. The weights are independent of the unweighted array, so under this conditioning the selected columns are iid Q samples. Thus their full helper Gram law is determined by the old weighted law. Lemma 5.3 applies and supplies the new uniform approximation property and affine Feller maps.

Preservation of the old constraints. The new exact maps preserve the actual old self-overlaps, since these are common deterministic values and the tilts are equivalent positive changes followed by finite row deletion. Intrinsicness gives commutation with the old finite-row maps. Proposition 4.5 gives commutation with the old Gaussian maps. Lemma 5.2 then restores the old marked identities with their original constants, including tests of the new centre coordinates, and preserves the old joint pair law.

The cavity objective is preserved as well. The normalization cocycle on disjoint fresh row blocks gives

$$(5.21) \quad \psi_A^{(\theta)}(T_U \rho) - \psi_A^{(\theta)}(\rho) = \psi_U^{(\theta)}(T_A \rho) - \psi_U^{(\theta)}(\rho) = 0,$$

because $T_A \rho = \rho$. The same identity holds for B , and therefore

$$(5.22) \quad P_\theta(T_U \rho) = P_\theta(\rho) = p_\theta.$$

For the Poisson factors, first condition on and truncate their degrees so that the cocycle involves bounded factors. The estimates

$$|\log \langle A \rangle - \log(2 \cosh h)| \leq cD, \quad |\log \langle B \rangle| \leq cK$$

pass to the untruncated factors. The fixed bounds on U control its normalization. Thus every new exact map preserves $\mathcal{D}$.

A common fixed point. Choose a countable uniformly dense family of positive continuous new probes, preserving a positive lower bound during approximation. The exact maps commute. For the first m maps average a law over their iterates indexed by $0 \leq k_1, \dots, k_m < L$. Applying any one of these maps changes the average in total variation by at most $2/L$. Compactness and continuity give a common fixed point for the first m maps. The nested compact fixed-point sets then give a common fixed point for the whole countable family. Uniform approximation of the factors extends stationarity to all positive continuous new probes. Denote this nonempty compact convex class by $\mathcal{D}'$.

Deterministic new self-overlaps. Choose an extreme point of $\mathcal{D}'$. For a bounded feature ϕ , finite-row stationarity passes to the empirical-mean tilt $e^{\lambda d_\phi(a)}$, where

$$d_\phi(a) = \lim_{L \rightarrow \infty} \frac{1}{L} \sum_{i=1}^L \phi^a(i),$$

by Lemma 4.2. For $\lambda > 0$ its change in directing mean is

$$(5.23) \quad \frac{Q(d_\phi e^{\theta \lambda d_\phi})}{Q(e^{\theta \lambda d_\phi})} - Q(d_\phi) = \frac{\text{Cov}_Q(d_\phi, e^{\theta \lambda d_\phi})}{Q(e^{\theta \lambda d_\phi})} \geq 0.$$

The covariance vanishes only when d_ϕ is Q -almost surely constant, as follows by integrating $(d - d')(e^{\theta \lambda d} - e^{\theta \lambda d'})/2$ over two independent copies. Stationarity forces its expectation to vanish.

Applying this to products of the new coordinates makes every required empirical product moment common to all columns, with value $D(Q)$.

The common value is invariant under every defining intrinsic map and every finite row permutation. Conditioning on a positive-probability event determined by $D(Q)$ therefore preserves the defining stationarities and row exchangeability. Lemma 5.2 preserves the old identities and pair constants under this conditioning. Both conditional laws belong to $\mathcal{S}_\theta$. Since P_θ is affine and globally minimized at p_θ , both parts of this decomposition must retain value p_θ . The conditional laws consequently belong to $\mathcal{D}'$. Extremality makes $D(Q)$ deterministic. The actual new self-overlap matrix, and hence the diagonals of its polynomial covariance probes, are now fixed.

New marked identities. Apply the Gaussian construction of Lemma 4.2 to the new probes. Its kernels have the deterministic diagonals just established. For each such covariance, the finite-deletion and conditional-tail argument in Lemma 5.1 gives a velocity $c_{K,\lambda}(Q)$ invariant under the defining intrinsic maps and finite row permutations. Conditioning on a level set preserves all old data by Lemma 5.2. The same global-minimum argument keeps the conditional law in $\mathcal{D}'$. Thus the conditioning hypothesis of Lemma 5.1 holds for this velocity inside the class actually constructed. Extremality makes it deterministic, and that lemma supplies all new generalized marked identities. Fixing the enlarged data. Freeze the enlarged actual self-overlaps and joint pair law attained at this extreme point. Intersect $\mathcal{D}'$ with these constraints and with all enlarged generalized marked identities using their now fixed pair constants. The conditions are affine because those constants are fixed. They are closed: empirical Gram entries have the uniform row approximation (5.18), and the new centre coordinates have the uniform finite-array approximations of Lemma 5.3. It suffices first to impose a countable continuous marked-cylinder determining class. Uniqueness of finite measures and a monotone-class argument then give all bounded measurable sampled-mark tests. The chosen extreme point belongs to the intersection, which is therefore nonempty, compact and convex. Every law in it retains (5.22). $\square$

5.5. Initialization.

Proposition 5.5. *For each $0 < \theta < 1$, there is a nonempty compact convex subclass $\mathcal{D}_0 \subseteq \mathcal{M}_\theta$ satisfying the assumptions of Theorem 5.4 for the initial physical feature list $\{1, x\}$ and the separate label-identity metadata.*

Proof. There are initially no old overlap constraints. Conditioning a law in $\mathcal{M}_\theta$ on an event measurable from Q and invariant under all raw cylinder maps and finite row permutations preserves $\mathcal{S}_\theta$. Affinity and global minimality then preserve the value p_θ in both conditional laws. Choose an extreme point of $\mathcal{M}_\theta$. The empirical-mean argument (5.23), applied first to $\{1, x\}$ and its products, fixes their actual self moments. Lemma 5.1 then gives the initial generalized marked identities, including the independent label-identity covariance.

Freeze this initial joint pair law and its self-overlaps and explicitly impose the initial marked identities with the fixed pair constants. Uniform empirical-row approximations make these closed affine constraints inside $\mathcal{M}_\theta$. The chosen extreme point proves that their intersection $\mathcal{D}_0$ is nonempty. Ordinary feature evaluations already satisfy the uniform finite-array approximation property, and finite-dimensional synchronization applies to their generalized identities. This proves the assertion. $\square$

For raw spins, $\{1, x\}$ spans the continuous one-site functions. For a general compact $X \subseteq [-1, 1]$, ordinary polynomials are uniformly dense and can be admitted through successive finite steps. Thus the construction starts with a finite list in either case. In the next section we organize the successive enrichment steps into one countable hierarchy, always admitting and fixing an operand before applying a centre operation to it.

6. A MASTER HIERARCHY AND ITS CENTRE ALGEBRA

The finite enrichment theorem allows us to add any prescribed finite collection of features and centres. We now organize these additions into one countable algebra and one scalar hierarchy. The hierarchy serves two purposes. Its sampling rule describes how a new replica can join an old family, and its geometry gives a common way to average the profiles in a cluster. We establish these facts before completing the algebra under centres of the final hierarchy.

Throughout this section, $0 < \theta < 1$ is fixed. A feature is a bounded one-site profile $\phi^a(i)$ attached to particle a . In the conditional iid representation of the site rows, we also write $\phi^a(v)$ and use $\langle \cdot, \cdot \rangle_v$ for the conditional $L^2(v)$ inner product. All statements about these profiles are made on their jointly row-exchangeable version.

6.1. Countable enrichment and a common scalar trace.

Proposition 6.1. *The minimizing class $\mathcal{M}_\theta$ contains a law ρ_* with a countable algebra $\mathcal{A}_0$ of bounded one-site features. The algebra contains 1 and the physical spin, is closed under rational polynomials, and contains the centres at all admitted cuts of every finite partial trace defined below. Every finite feature vector has deterministic actual self-overlaps and the full generalized marked Ghirlanda–Guerra identities.*

There is a scalar Gram array R such that these finite feature vectors synchronize to R . In particular, for $\phi, \psi \in \mathcal{A}_0$ there is a deterministic continuous function $K_{\phi\psi}$ on the support of the overlap law μ of R such that

$$(6.1) \quad \langle \phi^a, \psi^b \rangle_v = K_{\phi\psi}(R_{ab}).$$

The array R is ultrametric and satisfies the scalar Ghirlanda–Guerra identities with all bounded old sampled-mark tests. Its diagonal is 1, the event $R_{ab} = 1$ is exactly particle equality, and this terminal value is isolated from the off-diagonal support. Moreover,

$$(6.2) \quad \mu(\{1\}) = 1 - \theta.$$

Proof. Start with the finite list $\{1, x\}$ and the separate particle-identity metadata. Proposition 5.5 gives a nonempty compact subclass of $\mathcal{M}_\theta$ on which their actual self moments, joint pair law, and generalized marked identities are fixed. This is the initial class for Theorem 5.4.

Choose positive coefficients once and for all, for example

$$w_0 = \frac{1}{2}, \quad w_j = 2^{-j-1} \quad (j \geq 1).$$

Index zero is reserved for $K_0(a, b) = \mathbf{1}\{a = b\}$. Whenever a new scalar feature has been admitted and its moments have been fixed, assign it the next positive index. Enqueue rational polynomial operations, all finite-row probe arities, and the rational cuts and atom cuts of every finite partial trace. Atom cuts use included equality. A fair enumeration processes every operation after finitely many steps. For a composite expression, admit the operand and freeze its Gram data before taking its centre. Products evaluated at different sites enter as probes and covariance generators.

At each step Theorem 5.4 gives a nonempty compact subclass of the preceding class. The resulting nested classes have a nonempty intersection. Choose ρ_* in that intersection. The uniform finite-hull approximations in Lemma 5.3 make the relations defining each added feature closed on its fixed old-data class. Consequently every feature in the intersection remains its actual Borel function of the directing measure and the sampled particle. The initial list is sufficient for raw spins. If the same construction is made on a general compact $X \subset [-1, 1]$, the queued polynomial operations give the uniformly dense one-site algebra needed to approximate continuous functions on X .

For a scalar feature ϕ , choose a deterministic $B_\phi \geq \max(1, \|\phi\|_\infty)$ and put

$$\tilde{\phi} = 2 + \frac{\phi}{B_\phi}, \quad d_\phi = \|\tilde{\phi}^a\|_2^2, \quad \hat{\phi} = \frac{\tilde{\phi}}{\sqrt{d_\phi}}.$$

Here $d_\phi \in [1, 9]$ is deterministic and independent of a . Thus $\hat{\phi}$ has squared norm one, is positive, and is bounded by 3, uniformly over all admitted features. This affine change also preserves the coefficients minimizing a convex combination. Indeed, if m_ϕ is the common empirical mean and $t_a \geq 0$ with $\sum_a t_a = 1$, then

$$\left\| \sum_a t_a \tilde{\phi}^a \right\|_2^2 = 4 + \frac{4m_\phi}{B_\phi} + \frac{1}{B_\phi^2} \left\| \sum_a t_a \phi^a \right\|_2^2.$$

Let K_j be the Gram kernel of the normalized feature with index j . Positivity and Cauchy–Schwarz give $0 \leq K_j \leq 1$ and $K_j(a, a) = 1$. Set

$$(6.3) \quad R_{ab} = \sum_{j \geq 0} w_j K_j(a, b), \quad R_{ab}^{(L)} = \sum_{j \leq L} w_j K_j(a, b), \quad \varepsilon_L = \sum_{j > L} w_j.$$

The approximation is uniform, with $0 \leq R - R^{(L)} \leq \varepsilon_L$. The installed mixed covariance identities therefore pass to polynomial functions of R . Approximation on its compact range and a monotone-class argument give the full marked scalar identities. Scalar ultrametricity [3] applies to this Gram array.

To synchronize a fixed retained feature vector to this same R , take L large enough to contain its coordinates. Keep the coordinates $\sqrt{w_j} \hat{\phi}_j$, $1 \leq j \leq L$, in one common Hilbert summand. This retains their cross-feature blocks. Add the identity coordinate in an orthogonal summand and a tail coordinate with kernel

$$T_{ab} = \sum_{j > L} w_j K_j(a, b).$$

The resulting finite vector has trace exactly R and a deterministic full self-overlap matrix. It has a bounded row realization as follows. At each row choose a common channel $J > L$ with probabilities w_j/ε_L , an independent common Rademacher sign s , and independent particle-specific Rademacher signs η_a . Use the original site variable v for the retained coordinates and append

$$(6.4) \quad \sqrt{w_0} \eta_a, \quad \sqrt{\varepsilon_L} s \hat{\phi}_J^a(v).$$

Their kernels are $w_0 \mathbf{1}\{a = b\}$ and T_{ab} , and all their cross blocks with the retained coordinates vanish. Their absolute bounds are $\sqrt{w_0}$ and $3\sqrt{\varepsilon_L}$.

We check the generalized identities for this augmented vector. They involve products of quadratic forms in entrywise powers of its full overlap matrix, as in [7, Theorem 4 and equation (88)]. Each such quadratic form is a sum of a retained-block quadratic form, a constant multiple of T_{ab}^p , and a constant multiple of $\mathbf{1}\{a = b\}$. Powers of finite tail sums expand into products of old scalar kernels. Their products with retained quadratic forms are generated by the installed independent row-block probes. A product containing the identity kernel equals that kernel times deterministic self-block constants. Uniform convergence of the tail passes these identities to T . Polynomial approximation, uniqueness of finite measures, and a monotone-class argument give the bounded-measurable version of the generalized identities required by the synchronization theorem.

Finite-dimensional synchronization now applies. In particular, $K_j(a, b) = k_j(R_{ab})$ for deterministic nondecreasing functions k_j . Since the summands of the trace are nondecreasing,

$$(6.5) \quad |k_j(t) - k_j(s)| \leq \frac{|t - s|}{w_j}, \quad s, t \in \text{supp } \mu.$$

The same finite-vector argument synchronizes all retained cross blocks. Undoing the affine normalizations gives (6.1), including the blocks with the constant feature and the actual diagonal blocks.

Finally, $R_{aa} = 1$, whereas $R_{ab} \leq 1 - w_0$ for distinct labels. Thus its top event is particle equality. The weights have law PD($\theta, 0$), so its probability is $\mathbb{E} \sum_a v_a^2 = 1 - \theta$. This proves (6.2) and the terminal gap. $\square$

6.2. The scalar sampling rule. The next two lemmas concern a general scalar ultrametric array with marked Ghirlanda–Guerra identities. They apply whenever those hypotheses hold, independently of the construction in Proposition 6.1 and of any closure under centre operations.

For replicas $1, \dots, n$, let $\mathcal{F}_n$ be the sigma field generated by their old sampled marks and their Gram matrix. The scalar marked identities give, for $i \leq n$,

$$(6.6) \quad \text{Law}(R_{i,n+1} \mid \mathcal{F}_n) = \frac{1}{n}\mu + \frac{1}{n} \sum_{\substack{j \leq n \\ j \neq i}} \delta_{R_{ij}}.$$

The case $n = 1$ follows from the usual identities for $n \geq 2$. To see this, let f depend only on the first particle's marks and put

$$A = \mathbb{E}\langle f\psi(R_{12}) \rangle, \quad B = \mathbb{E}\langle f \rangle \int \psi \, d\mu.$$

Exchangeability of replicas 2 and 3, followed by the $n = 2$ identity, gives $A = (A + B)/2$, hence $A = B$.

Lemma 6.2. *Suppose a scalar ultrametric replica array satisfies (6.6) with all bounded old sampled-mark tests. The conditional law of the entire vector $(R_{i,n+1})_{i \leq n}$ given $\mathcal{F}_n$ depends only on the old Gram matrix. The same conclusion holds for any finite future Gram extension. After a finite coarsening of overlaps, the transition kernel depends only on the old coarsened Gram matrix.*

Proof. At a threshold t , partition the old replica indices into their $R \geq t$ classes. If C is one such class and $i \in C$, ultrametricity identifies the event that the new replica joins C with $R_{i,n+1} \geq t$. Its conditional probability is

$$(6.7) \quad \frac{\mu([t, \infty)) + |C| - 1}{n}.$$

As the threshold and the class vary, these events are nested or disjoint. Their probabilities therefore determine all finite Boolean combinations. A countable dense family of thresholds, together with the atoms of μ , determines the entire overlap vector. This proves the first assertion. Iterating the same kernel proves the assertion for future extensions. Restricting the argument to the finitely many coarsening thresholds proves the last assertion. Strict cuts can be included by monotone limits, with $\mu((t, \infty))$ in place of $\mu([t, \infty))$. $\square$

Lemma 6.3. *Every finite coarsening of a scalar ultrametric Ghirlanda–Guerra array has the genealogy law of the Poisson-product cascade with the corresponding cumulative masses. All finite canonical patterns at its nonempty levels have positive probability. At each nontrivial splitting level an occupied node has infinitely many children almost surely.*

Proof. We identify the sampling kernel directly. After omitting empty levels, write the cascade exponents as

$$0 < z_0 < \dots < z_{r-1} < 1, \quad z_{-1} = 0.$$

First consider a Poisson process on $(0, \infty)$ with intensity $ax^{-1-a} \, dx$, where $0 < a < 1$, and let S be the sum of its points. Its Laplace transform is

$$\mathbb{E}e^{-tS} = \exp\{-\Gamma(1-a)t^a\}.$$

Under the change of law proportional to S^b , with $0 \leq b < a$, the normalized points have law $\text{PD}(a, -b)$. For a specified partition of m samples into blocks of sizes $m_1, \dots, m_k$, its probability is

$$(6.8) \quad p_{a,-b}(m_1, \dots, m_k) = \frac{\prod_{j=1}^{k-1} (ja - b)}{(1-b)_{m-1}} \prod_{i=1}^k (1-a)_{m_i-1},$$

where $(x)_j = x(x+1)\dots(x+j-1)$ and $(x)_0 = 1$.

For completeness, insert

$$S^{-(m-b)} = \frac{1}{\Gamma(m-b)} \int_0^\infty t^{m-b-1} e^{-tS} dt$$

in the partition probability and apply the Poisson Campbell formula to the distinct selected points. Before normalization by $\mathbb{E}S^b$, the result is

$$\frac{a^k \prod_{i=1}^k \Gamma(m_i - a)}{\Gamma(m-b)} \int_0^\infty t^{ka-b-1} e^{-\Gamma(1-a)t^a} dt.$$

The normalizing moment is

$$\mathbb{E}S^b = \Gamma(1-a)^{b/a} \frac{\Gamma(1-b/a)}{\Gamma(1-b)},$$

with value 1 at $b = 0$. Substitution $u = \Gamma(1-a)t^a$ and cancellation give (6.8).

Stable Poisson marking gives the recursive cascade splitting. If a parent point u has an independent subtree total S' , replacing it by uS' preserves the parent exponent a and biases the subtree mark by $(S')^a$. The biased marks are independent of the transformed parent points. Recursing down the tree, the internal splitting at depth p has partition law (6.8) with $a = z_p$ and $b = z_{p-1}$.

Ratios of these partition probabilities show that, in a node containing m old samples, an occupied child of size m_i is selected with probability

$$\frac{m_i - a}{m - b},$$

and a new child is selected with probability

$$\frac{ka - b}{m - b}.$$

Multiplying along an occupied path telescopes. If C is an old class at depth $p+1$ and t is its threshold, its joining probability is

$$\frac{|C| - z_p}{n} = \frac{\mu([t, \infty)) + |C| - 1}{n}, \quad z_p = \mu((-\infty, t)).$$

These are exactly the probabilities in (6.7). Their nested class structure determines the entire coarse vector by Lemma 6.2, so induction on the number of replicas identifies the finite genealogy laws. The displayed probabilities for occupied and new children are positive at every admissible level, which proves positivity of finite patterns. The underlying stable Poisson process has infinitely many positive points. Its normalized splitting therefore has infinitely many children, and each is visited by an infinite iid sample with its positive mass. $\square$

6.3. Effective roots and the geometry of actual centres. We return to the law of Proposition 6.1. Included clusters are denoted by

$$C_q(a) = \{b : R_{ab} \geq q\}.$$

An *effective root* is a point of

$$(6.9) \quad S_+ = \{q : \mu([q, q + \delta)) > 0 \text{ for every } \delta > 0\},$$

together with the terminal value 1. All atoms and μ -almost every sampled branch height belong to this set. The points of $\text{supp } \mu$ excluded from it are nonatoms isolated on their right and form at most a countable set. At a numerical cut in a support gap, we use the actual root of the resulting cluster. Its centre norm is evaluated at that root.

For an admitted feature ϕ , let $c_q \phi^a$ be the actual minimum-norm point of the closed convex hull of its sampled profiles in $C_q(a)$, as in Lemma 4.3. This notation is available as a geometric definition before it is admitted as a feature in the completed algebra. At the terminal singleton, $c_1 \phi^a = \phi^a$.

Lemma 6.4. *Almost surely, every nonterminal branch height r observed between two Gibbs replicas has infinitely many strict $R > r$ children in its $R \geq r$ cluster.*

Proof. Retain the observed pair and reveal further ordinary Gibbs replicas. Among the first n indices, let m lie in its r -cluster and let $k \geq 2$ strict children be occupied, with multiplicities $m_1, \dots, m_k$. By (6.7) and its strict-cut version, the probability of entering the parent is

$$\frac{\mu([r, \infty)) + m - 1}{n},$$

and the probability of entering its j th occupied strict child is

$$\frac{\mu((r, \infty)) + m_j - 1}{n}.$$

The strict-child events are disjoint. Subtracting their probabilities from the parent probability gives the probability of discovering a new strict child,

$$(6.10) \quad \frac{(k-1)\mu((-\infty, r]) + \mu(\{r\})}{n} \geq \frac{\mu((-\infty, r])}{n}.$$

For almost every sampled r , the last numerator is positive. Conditionally on the initial observed Gram matrix, the probability of seeing no new child between times N and M is bounded by

$$\prod_{n=N}^M \left(1 - \frac{\mu((-\infty, r])}{n}\right),$$

which tends to zero. Repeat after each discovery and intersect the resulting countably many full-probability events. In particular, one can choose infinitely many representatives with exact mutual overlap r . $\square$

Lemma 6.5. *For every effective root q , every descendant a of a q -cluster, and every $\phi, \psi \in \mathcal{A}_0$,*

$$(6.11) \quad \langle \phi^a, c_q \psi^a \rangle_v = \langle c_q \phi^a, c_q \psi^a \rangle_v = K_{\phi\psi}(q).$$

For $s \leq t$ on the same lineage, interpreted as effective cluster roots,

$$(6.12) \quad \|c_t \phi^a - c_s \phi^a\|_2^2 = K_{\phi\phi}(t) - K_{\phi\phi}(s).$$

These identities hold simultaneously on all particle lineages. The centres are limits of representative averages chosen from the master genealogy, with approximation bounds uniform over descendants for each fixed feature.

Proof. At the terminal root the centre is the particle itself. Consider first a nonterminal cluster with infinitely many strict children of mutual overlap q . Choose one representative a_j from each child. The averages

$$A_m \phi = \frac{1}{m} \sum_{j=1}^m \phi^{a_j}$$

are Cauchy in $L^2(v)$. Their diagonal inner products are bounded and all off-diagonal inner products equal $K_{\phi\phi}(q)$. Their limit b_ϕ has squared norm $K_{\phi\phi}(q)$. Against a fixed descendant a , at most one representative lies in the same strict child. Thus

$$\langle \phi^a, b_\psi \rangle_v = K_{\phi\psi}(q), \quad \langle b_\phi, b_\psi \rangle_v = K_{\phi\psi}(q).$$

In particular, b_ϕ belongs to the closed convex hull and satisfies $\langle b_\phi, \phi^a - b_\phi \rangle_v = 0$ for every descendant. The projection characterization identifies it with the unique minimum-norm centre. Lemma 6.4 supplies these representatives at every sampled nonterminal branch height.

We next obtain the geometry simultaneously at effective roots. At an atom q , the finite coarsening using the parent cut $R \geq q$ and the child cut $R > q$ has a nontrivial splitting level. By Lemma 6.3, every occupied parent has infinitely many strict children. There are only countably many atoms. For

the remaining roots, fix countable cuts $u < b$ with $\mu([u, b)) > 0$. The finite-cut genealogy shows that every occupied u -cluster has infinitely many b -subclusters. Intersect these full-probability events over a countable dense family of cuts and over the countable particles. If $q \in S_+$ is a nonatom and $\delta > 0$, one can choose such $q < u < b < q + \delta$. The u -cluster of a descendant is contained in its q -cluster, so it supplies infinitely many representatives in distinct b -subclusters of that q -cluster. Their mutual overlaps lie in $[q, q + \delta)$.

Let B_ϕ bound $|\phi|$, and let $\omega_{\phi\psi}$ be a modulus of continuity of $K_{\phi\psi}$ on the master support. For an average of m representatives from these distinct subclusters, its inner product with any descendant satisfies

$$(6.13) \quad |\langle A_m \phi, \psi^a \rangle_v - K_{\phi\psi}(q)| \leq \omega_{\phi\psi}(\delta) + \frac{2B_\phi B_\psi}{m}.$$

The same bound controls the squared norm when $\phi = \psi$. By convexity, it also controls the inner product against any other representative average. Letting $m \rightarrow \infty$ and $\delta \downarrow 0$ therefore gives a Cauchy family. Its limit belongs to the closed hull and has the equal-inner-product projection identities above. This proves (6.11) at every effective root, with the asserted simultaneous version and uniform descendant bounds. The representatives depend only on the scalar genealogy and can be chosen for the whole countable feature list by diagonal refinement.

Finally, if $s \leq t$, every profile in the t -cluster has the same inner product with its ancestor centre $c_s \phi$. Passing to the convex-hull limit gives

$$\langle c_t \phi^a, c_s \phi^a \rangle_v = K_{\phi\phi}(s).$$

Expanding the squared difference proves (6.12). $\square$

6.4. Completing the algebra under master centres. The countable construction installed centres for partial traces. The following proposition identifies their limits as the centres of R and then justifies finite iterations of polynomial and centre operations. The order matters because the operand of a new centre must already have controlled profiles and a synchronized kernel.

Proposition 6.6. *For the law ρ_* , the centre of every admitted feature at an effective master root is a bounded L^2 limit of countably admitted partial-trace centre features. The approximation is uniform over descendants. The bounded completion of $\mathcal{A}_0$ is closed under finite polynomials and further centres at effective master roots, including finite iterations at sampled old branch heights.*

For each fixed numerical tuple of effective cut parameters, the completed features retain deterministic continuous synchronized kernels, actual self blocks, and their marked identities. Finite expressions evaluated at sampled old branch heights are available as old marked tests in the identities of the fixed master array R .

Proof. Approximation from partial traces. Uniform approximation in (6.3) gives

$$(6.14) \quad C_R(q) \subseteq C_{R(L)}(q - \varepsilon_L) \subseteq C_R(q - \varepsilon_L).$$

Choose $\delta_L \downarrow 0$ and a measurable rational-valued function

$$t_L(q) \in (q - \varepsilon_L - \delta_L, q - \varepsilon_L).$$

The centre at this admitted partial-trace cut satisfies the sharper inclusions

$$C_R(q) \subseteq C_{R(L)}(t_L(q)) \subseteq C_R(q - \varepsilon_L - \delta_L).$$

The effective root of the larger ancestor lies between $q - \varepsilon_L - \delta_L$ and q , unless the two cuts already define the same cluster across a gap.

For nested hulls, the projection inequality bounds the squared distance between their centres by the difference of their squared norms. Apply this inequality to the three displayed hulls, then use

(6.12) and (6.5). For the normalized feature with index j it gives

$$(6.15) \quad \sup_a \|c_{t_L(q)}^{(L)} \hat{\phi}_j^a - c_q \hat{\phi}_j^a\|_2^2 \leq \frac{\varepsilon_L + \delta_L}{w_j}.$$

Affine rescaling gives the corresponding estimate for each original feature. The use of cuts from below respects included equality at an atom. The estimate is deterministic on the simultaneous geometric event, so it also holds when q is a sampled branch height. The selection through $t_L(q)$ is countably valued and measurable.

Uniform completion of features. For a realized conditional row environment, put

$$\|\Phi\|_{\infty,2} = \sup_a \|\Phi^a\|_{L^2(v)}.$$

The preceding error estimate holds in this norm. On uniformly bounded features, a fixed polynomial is continuous in this norm: expand a product difference one factor at a time and apply the L^2 triangle inequality.

Fix an effective q -cluster. The representative averages in Lemma 6.5 are linear operators L_m chosen from its scalar subclusters. They can be chosen for the countable admitted list simultaneously, and they satisfy

$$(6.16) \quad \|L_m(\Phi - \Psi)\|_2 \leq \|\Phi - \Psi\|_{\infty,2}.$$

For an admitted feature their limit is its actual centre. Suppose now that uniformly bounded $\Phi^{(n)}$ converge to Φ in $\|\cdot\|_{\infty,2}$. Approximation through $\Phi^{(n)}$ and (6.16) show that $L_m\Phi$ is Cauchy. Its limit belongs to the closed convex hull of the Φ -profiles. The equal-inner-product projection identities pass from the old features through the uniform approximation, so this limit is that hull's actual minimum-norm centre. Taking limits in (6.16) gives

$$(6.17) \quad \|c_q\Phi^{(n)} - c_q\Phi\|_2 \leq \|\Phi^{(n)} - \Phi\|_{\infty,2}.$$

The construction applies in the countably many sampled clusters simultaneously. At the terminal singleton it is the identity. Thus the bound is uniform over descendants and proves closure under centres for this bounded completion.

Kernels and cut parameters. If all approximants and limits are bounded by B , then uniformly in a, b and in the row environment,

$$(6.18) \quad |\langle \Phi^a, \Psi^b \rangle_v - \langle \Phi^{(n),a}, \Psi^{(n),b} \rangle_v| \leq B\|\Phi - \Phi^{(n)}\|_{\infty,2} + B\|\Psi - \Psi^{(n)}\|_{\infty,2}.$$

For fixed numerical cut parameters the approximating kernels are deterministic continuous functions of R . Their uniform limits are therefore deterministic and continuous as well. An estimate first obtained almost everywhere under the pair law holds on its entire support by continuity. This includes actual diagonal blocks and blocks with 1, so all one-column moments remain deterministic. Bounded approximation passes the marked identities for fixed-parameter covariance kernels.

The geometric identities also pass to these completed features. In particular, for effective roots q, t ,

$$(6.19) \quad \|c_q\Phi - c_t\Phi\|_{\infty,2}^2 = |K_{\Phi\Phi}(q) - K_{\Phi\Phi}(t)|.$$

Together with (6.17) and polynomial continuity, this proves continuity of every finite expression in its effective cut parameters. When an intermediate operand also changes, insert and subtract its centre at the fixed cut before changing the cut itself. Induction through the finite expression then gives a common simultaneous version, first at countably many parameters and then at their effective limits.

To realize a finite iterated expression, approximate its first master-centre operands by admitted partial-trace centres and its polynomials by admitted polynomials. Apply (6.17) before approximating the next master-centre cut. Repeat through the expression. If a parameter is a previously sampled branch height, each rational-cut approximation is a measurable countably valued choice

of an admitted feature. Choose its required accuracy after its feature index is known. A finite iteration still has arbitrarily small prescribed error. Truncating these countably valued choices and using bounded convergence places the limiting expression among the old sampled-mark tests of the master identities.

The role of the parameters is consequently precise. At a fixed numerical tuple, a covariance kernel has its deterministic pair constant. Sampled old branch heights can be substituted into the simultaneous pathwise identities and into old marked tests. The subsequent branch recursion uses these tests with the fixed master kernel R . $\square$

7. CONDITIONAL INDEPENDENCE OF THE SITE ROWS

We now use the enriched hierarchy to determine the law of several site rows at once. The synchronized kernels describe inner products of one-state profiles. To pass from these pair data to a joint law, we successively replace the leaves of a deepest branch by their ancestor centres. Closure under iterated centre operations ensures that the observable obtained after each replacement is an available one-state feature. Appendix A gives a two-level example showing why repeated centre operations reveal information that a single centre layer can miss.

Fix $0 < \theta < 1$, and let ρ_* be the minimizing law constructed in Sections 5 and 6. Its physical message at particle a and site i is denoted by X_i^a . Write $\mathcal{A}$ for the completed algebra of bounded one-state features. Conditional row de Finetti gives a representation $\phi^a(v)$ of each feature, where the site variables v are independent under the conditional row law. The notation $\langle \cdot, \cdot \rangle_v$ refers to this conditional L^2 inner product. The conditional row environment itself may still be random.

For fixed numerical feature expressions $\phi, \psi \in \mathcal{A}$, Proposition 6.1 and Proposition 6.6 give deterministic continuous kernels satisfying

$$(7.1) \quad \langle \phi^a, \psi^b \rangle_v = K_{\phi\psi}(R_{ab}).$$

The diagonal value of the master is 1. Its isolated terminal class detects particle identity, and its overlap distribution μ has an atom of mass $1 - \theta$ there. The marked master identities allow bounded tests of the old sampled profiles, including the completed centre coordinates. We always use the actual sampled-hull centres $c_q\phi^a$ furnished by Proposition 6.6.

Theorem 7.1. *Sample particles $a^1, \dots, a^n$ with their Gibbs weights, and let $\Gamma_n = (R_{a^\ell a^{\ell'}})_{\ell, \ell' \leq n}$ be their master Gram matrix. There is a conditional one-row law κ_{Γ_n} such that, for every $d \geq 1$,*

$$(7.2) \quad \text{Law}(Y_1, \dots, Y_d \mid \Gamma_n) = \kappa_{\Gamma_n}^{\otimes d}, \quad Y_i = (X_i^{a^1}, \dots, X_i^{a^n}).$$

The same assertion holds jointly for any finite collection of enriched feature coordinates. Their conditional moments are obtained by the deepest-branch recursion proved below.

7.1. Tensor centres and ancestor conditional expectations. The centre geometry from Lemma 6.5 applies to each fixed feature before independence of the site rows is known. We first record its consequence for products at independently sampled sites.

Lemma 7.2. *Let q be an effective root and let $\phi_1, \dots, \phi_k \in \mathcal{A}$. In the conditional space $L^2(v_1, \dots, v_k)$, the centre of the profiles*

$$a \longmapsto \prod_{i=1}^k \phi_i^a(v_i), \quad a \in C_q,$$

is

$$(7.3) \quad \prod_{i=1}^k c_q \phi_i(v_i).$$

Proof. At the terminal root the cluster contains one particle, which gives the assertion immediately. At a nonterminal root use the representative averages constructed in Lemma 6.5. When the representatives have exact mutual overlap q , the inner product of two distinct tensor profiles is

$$\prod_{i=1}^k K_{\phi_i \phi_i}(q).$$

Their averages are therefore Cauchy. Let B denote the limit and put $b = \prod_i c_q \phi_i(v_i)$. The mixed centre identities in that lemma give

$$\|B\|_2^2 = \|b\|_2^2 = \langle B, b \rangle = \prod_{i=1}^k K_{\phi_i \phi_i}(q).$$

Hence $B = b$. The same identities show that every descendant tensor profile has inner product $\|B\|_2^2$ with B . As B belongs to the closed convex hull, this is its minimum-norm projection identity.

At a nonisolated effective root take representatives in distinct $(q + \delta)$ -subclusters, as in Lemma 6.5. All cross inner products converge to the displayed product as $\delta \downarrow 0$. Boundedness and continuity of the finitely many kernels pass the norm and mixed inner-product identities to the limit. This proves the assertion at every effective root. $\square$

For a particle a , let $Z_q^a(v)$ collect the one-site ancestor-centre coordinates on its lineage at effective heights at most q . Include all coordinates $c_q \phi^a(v)$ in this collection. These coordinates are common to every particle of the q -cluster.

Lemma 7.3. *In each realized conditional row environment,*

$$(7.4) \quad c_q \phi^a = \mathbb{E}_v[\phi^a \mid \sigma(Z_q^a)].$$

Consequently, if h is a bounded function of these ancestor coordinates in the bounded closure of their polynomial algebra, then

$$(7.5) \quad c_q(h\phi)^a = h c_q \phi^a.$$

The corresponding conditional expectations satisfy the tower identity along each lineage.

Proof. A polynomial $h(Z_q)$ is constant as a profile across the q -cluster. Thus $c_q h = h$. Applying the mixed centre identity of Lemma 6.5 to ϕ and h gives

$$\langle \phi^a - c_q \phi^a, h(Z_q^a) \rangle_v = 0.$$

The variable $c_q \phi^a$ is measurable with respect to $\sigma(Z_q^a)$. Bounded polynomial cylinder functions are dense in the resulting L^2 space, so this orthogonality identifies the conditional expectation. The module and tower identities now follow from the corresponding properties of conditional expectation. $\square$

The simultaneous centre versions from Proposition 6.6 allow a sampled branch height r in these formulas. More explicitly, choose the installed cuts approaching r from below. The effective roots of their ancestor clusters either equal r or converge to r from below. The increment identity

$$\|c_t \phi^a - c_s \phi^a\|_2^2 = K_{\phi\phi}(t) - K_{\phi\phi}(s)$$

gives convergence uniformly over descendants for each fixed feature. Products of bounded features preserve it. Countably valued choices of the approximating cuts can be truncated and passed through the marked identities. We use this version whenever a centre in the recursion below is evaluated at an old random branch height.

7.2. Extending a deepest branch at its exact height. Lemma 6.2 gives the following conditional kernel for every old index $i \leq n$:

$$(7.6) \quad \text{Law}(R_{i,n+1} \mid \text{all old sampled marks}) = \frac{1}{n}\mu + \frac{1}{n} \sum_{\substack{j \leq n \\ j \neq i}} \delta_{R_{ij}}.$$

The conditional law of the entire new-overlap vector, and of every finite future Gram extension, depends only on the retained Gram matrix. This remains true after a finite coarsening. In particular, conditioning on a future Gram extension does not bias the retained marked law once the retained Gram has been fixed.

First identify any repeated terminal labels, multiplying their observable factors. The preceding property permits deletion of the duplicate indices while keeping the marked law conditional on the reduced Gram. We can therefore work with distinct particles.

A deepest branch of their finite induced tree consists of $b \geq 2$ occupied children that are single leaves $a_1, \dots, a_b$. Their mutual overlap is a nonterminal value r . Every outside particle has the same overlap with all these leaves, and that overlap is strictly less than r . We call this collection of leaves a star. In the next lemma n counts the distinct retained replicas, including the outside particles.

Lemma 7.4. *Conditional on the old marked configuration, a new replica has overlap exactly r with every star leaf and the prescribed overlaps with every outside particle with probability*

$$(7.7) \quad p_{n,b}(r) = \frac{(b-1)\mu((-\infty, r]) + \mu(\{r\})}{n}.$$

This is positive for almost every sampled nonterminal branch height. Repeated conditioning on this extension produces an infinite exchangeable star while retaining all outside marks and all common ancestor-centre coordinates.

Proof. By the class-membership form of Lemma 6.2, the new replica enters the parent $R \geq r$ class with probability

$$\frac{\mu([r, \infty)) + b - 1}{n}.$$

Each occupied strict $R > r$ child contains one old leaf, so the probability of entering that child is $\mu((r, \infty))/n$. Subtracting these b disjoint events gives

$$\frac{\mu([r, \infty)) + b - 1 - b\mu((r, \infty))}{n} = p_{n,b}(r).$$

Ultrametricity then forces all outside overlaps. Positivity holds whenever $\mu((-\infty, r]) > 0$ or $\mu(\{r\}) > 0$, which excludes at most a null bottom endpoint among sampled branch heights.

We describe the conditional versions needed for iteration. Let H record the outside Gram and the overlaps between the outside particles and the star. Disintegrate the original marked law at the parameters (H, r) and the prescribed b -star pattern, and denote its kernel by $L_b(H, r)$. Conditioning on the next extension multiplies the parameter marginal by the strictly positive factor $p_{n,b}(r)$. The old and new parameter marginals are therefore equivalent. Since this factor depends only on the Gram data, the old-mark marginal of the extended conditional law remains $L_b(H, r)$.

Choose common versions outside the union of the null sets for the countably many extension lengths. For these versions, restriction of $L_{b+1}(H, r)$ to its first b star leaves is $L_b(H, r)$. Replica symmetry gives invariance under every permutation of the star leaves, and hence consistency for deletion of any leaf. Kolmogorov extension gives an infinite exchangeable star. The outside marked array and the common centre coordinates have been retained in every kernel, so the exchangeability holds jointly with these variables and conditionally on them.

This disintegration also applies when r has a continuous law. The initial sampled pair already supplies its value. Each subsequent exact-height extension then has the positive conditional probability computed above. $\square$

7.3. A conditional square and the law of a star. The tensor-centre identity now determines the directing law of the exchangeable star. For clarity, all site coordinates in the next argument are the actual evaluations in the jointly row-exchangeable law.

Lemma 7.5. *In the infinite star of Lemma 7.4, conditional on the common ancestor coordinates Z_r , the star particles are independent and identically distributed and are independent of all outside marks. At finitely many sites their conditional law is the product of its one-site laws. The one-site moments are characterized by $c_r\phi$ for $\phi \in \mathcal{A}$.*

Proof. Fix sites $v_1, \dots, v_k$ and bounded features $\phi_1, \dots, \phi_k$. For a star particle a put

$$F_a = \prod_{i=1}^k \phi_i^a(v_i), \quad c_F = \prod_{i=1}^k c_r \phi_i(v_i).$$

Let Γ denote the prescribed finite genealogy parameters. For two distinct star leaves, the tensor-centre identities imply

$$(7.8) \quad \mathbb{E}[(F_{a_1} - c_F)(F_{a_2} - c_F) \mid \Gamma] = 0.$$

Indeed, after averaging the site coordinates, all four terms in the expansion equal $\prod_i K_{\phi_i \phi_i}(r)$. Conditioning on Γ preserves row exchangeability and does not distinguish the indicated site coordinates. Thus these expectations are precisely the conditional L^2 inner products of (7.1) and Lemma 7.2.

Apply conditional de Finetti to the star, retaining the outside marked array and the common Z_r coordinates at all observed sites. Let M_F be the random directing moment of F_a . Conditional independence under the directing law, together with the measurability of c_F from Z_r , rewrites the left side of (7.8) as

$$(7.9) \quad \mathbb{E}[(M_F - c_F)^2 \mid \Gamma].$$

It follows that $M_F = c_F$ almost surely, including after conditioning on the outside marks. Use a countable polynomial determining class of finite feature blocks to make these identities simultaneous. The directing law is then determined by Z_r alone, and its moments factor over the sites. Moment determinacy on bounded coordinate products proves the stated conditional independence assertions. $\square$

7.4. Collapsing the finite genealogy.

Proof of Theorem 7.1. Fix a finite sampled Gram matrix and a product of one-state feature evaluations at finitely many sites. After combining repeated particles, choose a deepest star with leaves $a_1, \dots, a_b$ and height r . At each site, Lemma 7.5 replaces its leaf factors by

$$(7.10) \quad \prod_{j=1}^b c_r \phi_j.$$

Attach this one-state feature to one representative leaf and delete the other leaves. It is a polynomial of centre features in the completed algebra. At a later branch we may therefore take its centre again.

The height r may disappear from the retained Gram. Keep it as an external parameter, along with the entire deleted Gram extension. By Lemma 6.2, the conditional distribution of that extension given all retained marks depends only on the retained Gram. Disintegration therefore leaves the retained marked law unchanged when the deleted parameters are specified. At every subsequent step we retain all previously deleted heights, and the same iterated extension property applies. Thus every current observable is a bounded polynomial of features of a single retained particle, with the deleted heights as numerical parameters. Proposition 6.6 supplies the simultaneous measurable versions needed when these parameters have continuous distributions. A later centre is taken on the current particle's lineage and does not require a new branching event at a deleted height.

There are at most $n - 1$ collapses for n distinct particles. At the final particle, every one-site polynomial moment of the completed feature vector is deterministic. For a feature ϕ this follows from the block with the constant feature:

$$(7.11) \quad \mathbb{E}_v \phi^a(v) = \langle \phi^a, 1 \rangle_v = K_{\phi,1}(1).$$

The same identity applies to all polynomials generated in the recursion. Conditional row de Finetti and moment determinacy on the compact products of bounded feature coordinates give a deterministic law for one row of this particle. Its rows are therefore independent and identically distributed. When the feature expressions contain deleted heights, this statement is applied after disintegration with those numerical parameters fixed.

Working backwards through (7.10) factorizes the original moments over the sites, conditional on the original finite Gram matrix. Product tests determine the joint row law, which proves (7.2) and the assertion for enriched feature coordinates. $\square$

8. RECOVERY BY FINITE CASCADES

The conditional law in Theorem 7.1 is indexed by the full sampled master Gram matrix. We approximate this law on a sequence of finite coarse genealogies. At each resolution, the conditional law of one row is consistent across finite replica samples. A hierarchical representation of that one-row law, followed by independent copies across sites, gives a finite cascade trial. Convergence to the original law will be proved for complete finite rectangles of replicas and sites.

Proposition 8.1. *Fix $0 < \theta < 1$, and let ρ_* be the enriched minimizing law constructed above. Its physical cavity value p_θ is a limit of values of finite cascade trials with independent site marks and a possible site-root mark. Their terminal mass exponent can be kept equal to θ . Removing the root mark by an additional level of exponent tending to zero gives*

$$(8.1) \quad \mathcal{P}_* \leq p_\theta.$$

8.1. Consistent laws on a finite coarse tree. Choose nested finite partitions of the nonterminal master support whose mesh tends to zero. Retain only intervals of positive μ -mass and keep the isolated terminal singleton $\{1\}$ in every partition. For k Gibbs replicas, denote the full master Gram matrix by Γ_k and its j th coarse pattern by Γ_k^j . Then

$$(8.2) \quad \sigma(\Gamma_k^j) \uparrow \sigma(\Gamma_k).$$

For each finite coarse pattern T , let $\kappa_{j,T}$ be the conditional law of one physical-message row of the original replicas given $\Gamma_k^j = T$. Every admissible canonical pattern has positive probability by Lemma 6.3. Indeed, its genealogy is that of the finite cascade with the corresponding cumulative masses, and the nonempty levels have strictly positive existing-child and new-child sampling probabilities. Repeated terminal labels force equality of the corresponding row coordinates, because the terminal singleton detects particle identity.

The coarsened sampling kernel from Lemma 6.2 gives the consistency relation

$$(8.3) \quad (\kappa_{j,T'})|_{\text{old leaves}} = \kappa_{j,T} \quad \text{whenever } T' \text{ extends } T.$$

To see this, the conditional probability of a future coarse Gram extension, given all old marks and the old fine Gram, is a function of the old coarse pattern alone. Conditioning on the extension therefore leaves every old row observable unchanged once that pattern is fixed. Replica symmetry also makes $\kappa_{j,T}$ invariant under permutations that preserve T .

Take a countably branching tree of the finite depth r_j specified by the partition. For every finite list of canonical leaves prescribe the law $\kappa_{j,T}$ corresponding to its genealogical pattern. Equation (8.3), together with the rule for repeated labels, gives Kolmogorov consistency. The resulting single-row field on $\mathbb{N}^{r_j}$ is hierarchically exchangeable.

The one-tree representation of [2, Theorem 1, equation (7)] applies to this standard Borel field. It gives a deterministic measurable function f_j such that

$$(8.4) \quad m_a = f_j(U_*, U_{a|1}, \dots, U_{a|r_j}), \quad a \in \mathbb{N}^{r_j},$$

where the uniforms at all vertices, including the root, are independent. The representation can also be read recursively. The child subarrays at the root form an exchangeable sequence. Their directing law is almost surely invariant under the hierarchical permutations inside a child. Applying the same observation at each remaining level and realizing the resulting standard Borel kernels by uniforms yields (8.4).

Use an independent copy of this entire family for each physical site, including its root uniform. Give the field independent cascade weights with the coarse genealogy law. Conditional on any sampled list of canonical leaves, the one-row law is $\kappa_{j,T}$, and it depends on the list only through its coarse pattern T . The independent weights give exactly the original distribution of that pattern. Thus the new trial has, conditional on T , independent rows with common law $\kappa_{j,T}$. The retained terminal atom has mass $1 - \theta$, so the last cascade exponent is θ . Omitting empty levels makes all previous exponents strictly increasing in $(0, \theta)$.

This construction imposes row independence at the coarse level on the new trial. Its relation to the original conditional law follows from the limit below.

8.2. Convergence of the replicated message arrays. Write κ_{Γ_k} for the conditional one-row law of the original sample given its full Gram matrix. For a bounded one-row test g on k replicas, conditional expectation gives

$$(8.5) \quad \kappa_{j, \Gamma_k^j}(g) = \mathbb{E}[\kappa_{\Gamma_k}(g) \mid \Gamma_k^j] \longrightarrow \kappa_{\Gamma_k}(g) \quad \text{in } L^1.$$

The convergence follows from (8.2) and the martingale convergence theorem.

For bounded row tests $g_1, \dots, g_d$, expand the difference of the two products by replacing one factor at a time. Uniform boundedness and (8.5) yield

$$(8.6) \quad \mathbb{E} \prod_{i=1}^d \kappa_{j, \Gamma_k^j}(g_i) \longrightarrow \mathbb{E} \prod_{i=1}^d \kappa_{\Gamma_k}(g_i).$$

The left side is the corresponding replicated-array expectation of the finite trial. By Theorem 7.1, the right side is the expectation for the original enriched law. Product tests determine the finite rectangular laws on the compact message space. Therefore every fixed collection of replicas and sites in the finite trials converges jointly to the corresponding collection in ρ_* .

8.3. The logarithmic functional and the root mark.

Proof of Proposition 8.1. We apply Proposition 2.7 to the finite replicated-array convergence just proved. We spell out the finite approximations because the functional contains both unbounded Poisson degrees and a logarithm.

Recall $c = \beta J_0$. Lemma 2.1 gives

$$(8.7) \quad 2 \cosh h e^{-cD} \leq A_a \leq 2 \cosh h e^{cD}, \quad e^{-cK} \leq B_a \leq e^{cK}.$$

First retain only degrees at most L . The corresponding positive weighted integrals lie in deterministic compact intervals bounded away from zero. Approximate the logarithm uniformly on those intervals by a polynomial. A power of a weighted integral is an expectation over finitely many Gibbs replicas. With the degrees bounded, each factor is a continuous multilinear function of finitely many message coordinates. Hence (8.6) passes every term of the polynomial approximation, and then the approximation error can be sent to zero.

Finally, (8.7) implies

$$|\log \langle A \rangle - \log(2 \cosh h)| \leq cD, \quad |\log \langle B \rangle| \leq cK.$$

The omitted tails are bounded uniformly by

$$c\mathbb{E}[D\mathbf{1}_{\{D>L\}}] + c\mathbb{E}[K\mathbf{1}_{\{K>L\}}] \longrightarrow 0.$$

Thus the full cavity values of the finite trials with a root mark converge to $P_\theta(\rho_*) = p_\theta$ at this fixed θ .

The root uniform in (8.4) is independent for different physical sites. Proposition 2.5 incorporates it as a nonroot mark by prepending a cascade level of exponent $\eta > 0$, chosen below the first existing exponent. Its uniform estimate is

$$(8.8) \quad |\mathcal{P}_\eta - \mathcal{P}^{\text{root}}| \leq C_{\text{root}}\eta, \quad C_{\text{root}} = \frac{c^2}{2}(5\alpha^2 + 3\alpha).$$

Here $5\alpha^2 + 3\alpha = \mathbb{E}D^2 + \mathbb{E}K^2$. Sending $\eta \downarrow 0$ for each finite trial puts its value in the closure of the class (1.5). This proves (8.1). $\square$

Remark 8.2. The deepest-branch recursion also describes conditional moments directly on finite tree patterns. It is a finite expression in multiplication and centre operators. For a fixed feature, the increment identity of Lemma 6.5 controls a change in an effective root height. For a changing intermediate feature, insert and subtract its limiting feature and use the uniform-over-descendants contraction from Proposition 6.6. When branch heights merge, the module identity (7.5) identifies the limiting expression with the merged-branch expression. These facts give continuous conditional moment functions on finite patterns with effective support-root heights, including limits of patterns from positive-probability windows. The martingale argument above needs no modulus uniform in the number of levels.

8.4. Completion of the variational formula.

Proof of Theorem 1.1. Propositions 2.4 and 2.3 give the finite thermodynamic limit $F = \sup_N F_N$ and the bound $F \leq \mathcal{P}_*$. Fix $0 < \theta < 1$. Proposition 3.1 supplies an ordinary stationary spin law with cavity value F , and Proposition 3.4 gives

$$p_\theta \leq F + \frac{2\alpha c(1 - \theta)}{\theta}.$$

Theorem 5.4 and Proposition 6.1 construct an enriched law in the global minimizing class. The scalar genealogy and synchronized centre geometry are established before Proposition 6.6 completes its centre algebra. Theorem 7.1 therefore applies to that law. Proposition 8.1 now gives

$$(8.9) \quad F \leq \mathcal{P}_* \leq p_\theta \leq F + \frac{2\alpha c(1 - \theta)}{\theta}.$$

Letting $\theta \uparrow 1$ proves the formula. $\square$

The construction also provides an explicit order for a diagonal sequence of trials. For $n \geq 2$, set $\theta_n = 1 - 1/n$ and first complete the compact countable enrichment at this fixed exponent. Choose a finite coarse trial with root mark whose complete cavity value differs from p_{θ_n} by at most $1/n$. The degree cutoff and logarithm-polynomial approximation in the preceding proof are chosen before the required finite-array accuracy. For the resulting trial, let $\zeta_0^{(n)} > 0$ be its first exponent and choose

$$0 < \eta_n < \min \left\{ \zeta_0^{(n)}, \frac{1}{n(1 + C_{\text{root}})} \right\}.$$

After prepending this level, the trial is admissible in (1.5) and has value $\mathcal{P}_n$ satisfying

$$(8.10) \quad F \leq \mathcal{P}_n \leq F + \frac{2\alpha c}{n-1} + \frac{2}{n}.$$

Its depth may grow with n . All constructions and approximations have been completed at fixed θ_n before the exponent tends to one.

If $\alpha = 0$, $\beta = 0$, or $\nu = \delta_0$, the cavity factors reduce directly to $A_a = 2 \cosh h$ and $B_a = 1$. In these cases $F = \mathcal{P}_* = \log(2 \cosh h)$. The proof otherwise allows every real h and every bounded symmetric coupling law in the model.

APPENDIX A. WHY THE CENTRE ALGEBRA MUST BE ITERATED

The collapse in Section 7 produces polynomials of ancestor centres, whose centres are needed at the next branch. The following two-level construction illustrates the information revealed by that second centre operation. All pair blocks of the raw spins and their first centre layer can be deterministic while a joint moment of several particles still retains a common random parameter.

Let W be a fair Bernoulli variable and define two probability laws on $[0, 1]$ by

$$\lambda_0 = \frac{1}{2}\delta_{1/4} + \frac{1}{2}\delta_{3/4}, \quad \lambda_1 = \frac{4}{5}\delta_{3/8} + \frac{1}{5}\delta_1.$$

Conditional on W , let $(T_i)_{i \geq 1}$ be independent with law λ_W . At each site i , independently for the parent labels $a \in \mathbb{N}$ conditional on T_i , choose

$$\mathbb{P}(\Theta_{a,i} = 1 \mid T_i) = \mathbb{P}(\Theta_{a,i} = -1 \mid T_i) = T_i/2, \quad \mathbb{P}(\Theta_{a,i} = 0 \mid T_i) = 1 - T_i.$$

For each child $b \in \mathbb{N}$, independently conditional on the parent variables, choose $X_{ab,i} \in \{-1, 1\}$ with

$$\mathbb{E}[X_{ab,i} \mid \Theta_{a,i}] = \Theta_{a,i}.$$

All choices at different sites are independent conditional on W . Independent two-level cascade weights may be placed on the leaves (a, b) .

The first two moments of the propensity laws agree:

$$(A.1) \quad \int t \, d\lambda_w(t) = \frac{1}{2}, \quad \int t^2 \, d\lambda_w(t) = \frac{5}{16}, \quad w \in \{0, 1\}.$$

Their third moments are

$$(A.2) \quad \int t^3 \, d\lambda_0(t) = \frac{7}{32}, \quad \int t^3 \, d\lambda_1(t) = \frac{31}{128}.$$

Consequently, the conditional empirical Gram matrix of the spin profiles is

$$\langle X_{ab}, X_{a'b'} \rangle_v = \begin{cases} 1, & (a, b) = (a', b'), \\ 1/2, & a = a', \, b \neq b', \\ 0, & a \neq a'. \end{cases}$$

The parent-cluster centre of X_{ab} is Θ_a . Averaging distinct child profiles gives this limit, and its inner product with every child equals its squared norm $1/2$. The root centre of X is zero.

The algebra of one-state polynomials in (X, Θ) is spanned by $1, X, \Theta, \Theta^2$, because

$$X^2 = 1, \quad \Theta^3 = \Theta, \quad X\Theta = \Theta^2, \quad X\Theta^2 = \Theta.$$

Its complete pair blocks within a parent depend only on $\mathbb{E}T = 1/2$. The pair blocks between different parents depend only on $\mathbb{E}T$ and $\mathbb{E}T^2 = 5/16$. For instance,

$$\langle \Theta_a^2, \Theta_{a'}^2 \rangle_v = \begin{cases} 1/2, & a = a', \\ 5/16, & a \neq a'. \end{cases}$$

Thus all these pair blocks and all polynomial one-column moments have the same values for $W = 0$ and $W = 1$.

To see the remaining dependence, choose two distinct children in each of three distinct parents and set

$$Y_i = \prod_{a=1}^3 X_{a1,i} X_{a2,i}.$$

Conditional independence of the children and then of the parents given T_i gives

$$\mathbb{E}[Y_i \mid W] = \mathbb{E}[\Theta_{1,i}^2 \Theta_{2,i}^2 \Theta_{3,i}^2 \mid W] = \int t^3 d\lambda_W(t).$$

The two possible means differ by $3/128$. For distinct sites i, j , conditioning on the six-leaf genealogy therefore gives

$$(A.3) \quad \text{Cov}(Y_i, Y_j \mid \text{six-leaf genealogy}) = \text{Var}(\mathbb{E}[Y_i \mid W]) = \frac{9}{65536} > 0.$$

The same calculation holds for leaves sampled using the independent cascade weights and conditioned to have this pattern.

The next centre operation reveals W through a one-column moment. The root centre of the parent feature Θ_a^2 is the profile T : averages over distinct parents converge to T in conditional $L^2(v)$, and

$$\langle \Theta_a^2, T \rangle_v = \langle T, T \rangle_v = 5/16$$

verifies the projection identity. Once this centre and its polynomials belong to the algebra,

$$\langle [c_{\text{root}}(\Theta^2)]^3, 1 \rangle_v = \int t^3 d\lambda_W(t)$$

is random. The deterministic self-block property of the fully iterated centre algebra excludes this construction. This is the property used at the last particle in (7.11).

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